FSDFormer: A Frequency-Selected Differential Fusion Transformer for Remote Sensing Image Spatiotemporal Fusion

Sichen Lu, Juanjuan Jing, Lei Yang, Boyang Nie, Lei Feng, Xiaoying He, and Jinsong Zhou

***Abstract*—Spatiotemporal fusion (STF) plays a critical role in remote sensing and Earth observation by integrating high spatial resolution details with high temporal consistency, serving as an essential tool for environmental monitoring and postdisaster assessment. Recently, numerous STF methods, especially deep learning-based approaches, have been proposed and shown promising results. However, most of them struggle with effectively capturing complex spatiotemporal interactions as well as balancing computational efficiency and interpretability. To address these challenges, we propose an innovative STF framework, frequency-selected differential fusion Transformer (FSDFormer), which comprehensively processes multiscale spatiotemporal data while maintaining linear computational complexity. FSDFormer integrates three key components: the spatial enhanced agent Transformer (SEAT) block, the frequency-adaptive filtering network (FAFN), and the spatiotemporal differential fusion (SDF) block. The SEAT block employs a novel attention mechanism to reduce computational complexity, while the FAFN enhances feature extraction by introducing frequency-domain filtering in the forward network. Grounded in a spatiotemporal dependency-aware fusion strategy, the SDF block leverages a two-stage mechanism to excavate and fuse temporal and spatial information sequentially. Extensive experiments show that FSDFormer delivers superior performance on three STF datasets, outperforming state-of-the-art (SOTA) methods in accuracy, robustness, and generalizability. The code is available at https://github.com/Sichen-Lu/FSDFormer**

***Index Terms*—Computational efficiency, deep learning, frequency selection, remote sensing, spatiotemporal fusion (STF), Transformer.**

I. Introduction

REMOTE sensing imagery with high spatial and temporal resolution is essential for understanding and monitoring of land-cover dynamics [1], [2]. These images provide extensive characterization of surface features [3], [4], allowing for accurate investigation of environmental changes [5], agricultural activities [6], [7], urban development [8], [9], and phenological analysis [10]. However, the tradeoff between spatial resolution and rapid response in satellite sensor design restricts the simultaneous achievement of high spatial and temporal acuity in Earth observation systems [11]. One typical imaging platform is Landsat satellites, such as Landsat 8 and Landsat 9, which have a higher spatial resolution of 15–30 m but have a revisit period of approximately 16 days. In contrast, the Moderate Resolution Imaging Spectroradiometer (MODIS) records data with spatial resolutions ranging from 250 to 1000 m and a revisit duration of 1–2 days. Unlike early bandwise fusion strategies [12], [13], [14], spatiotemporal fusion (STF) combines data from several sensors or observations on different dates, taking advantage of high resolution in both spatial and temporal domains [15]. Such developments have the potential to considerably increase the value of remote sensing imaging in dynamic monitoring and decision-making across a wide range of applications.

Many strategies have been developed in STF over the last few decades and can be divided into five categories [16]: weight function-based, unmixing-based, Bayesian-based, hybrid, and deep learning-based methods. The first four categories were primarily used by early STF frameworks to take advantage of spatial–temporal correlations in remote sensing data. Despite emphasizing theoretical interpretability, the output quality of many conventional algorithms is often reduced due to their susceptibility to nonlinear land-cover changes, multispectral resolution mismatches, and registration errors [17], especially in heterogeneous regions [18]. Additionally, the pixel-centric processing paradigm and manual feature engineering of these methods restrict their scalability and generalization potential in extensive spatiotemporal applications.

While deep learning methods are excellent at modeling complex nonlinear relationships and automatically extracting prior information, they still have a number of serious drawbacks [19]. Many algorithms have trouble striking a balance between temporal and spatial consistency, which often leads to artifacts or the loss of important details in the fused output. Some approaches are computationally intensive, posing challenges for practical large-scale deployment [20]. Furthermore, the interpretability of fusion processes remains a major

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challenge, since reliance on black-box models hinders alignment with spatiotemporal dependency-aware techniques and obscures insights into the fusion mechanism. The contradiction between traditional and learning-based approaches reveals a crucial research gap: how to balance accuracy, robustness, and transparency by combining interpretable physical models and data-driven learning. Addressing this gap could open the way for next-generation STF frameworks capable of handling the growing complexity of remote sensing data integration.

To overcome these issues, we introduce an innovative STF approach called the frequency-selected differential fusion Transformer (FSDFormer), designed for efficient and interpretable fusion of features. FSDFormer employs a U-shaped hierarchical encoding–decoding architecture, capturing multiscale information spanning local to global scales, which is essential for image fusion. Initially, FSDFormer extracts features separately from a fine image and two temporally adjacent coarse images using shared-weight encoders to maintain temporal consistency. In the model's backbone, we introduce an agent attention mechanism that leverages redundancy in attention weights to enhance expressive power while reducing computational complexity. Additionally, a frequency-adaptive filtering network (FAFN) is integrated into the feed-forward network (FFN), adaptively filtering the most informative frequency signals to improve high-frequency detail capture.

During the fusion phase, we modify the agent's attention calculation and incorporate an increment-based model to predict high spatial resolution differential information, thereby enhancing interpretability and aligning more closely with the spatiotemporal dependency-aware fusion approach. These breakthroughs enable FSDFormer to achieve efficient computation while enhancing both the accuracy and interpretability of fusion results, outperforming state-of-the-art (SOTA) methods.

In summary, we proposed FSDFormer, a pioneering STF method that overcomes the major limitations of previous fusion approaches by a unique architectural design. The key contributions of this work are summarized in the following.

1) *Novel STF Framework for Efficient Multiscale Fusion:* Unlike conventional Transformer-based approaches, which have quadratic complexity, our framework includes an agent attention mechanism that preserves multiscale feature sensitivity while achieving linear computational efficiency. Furthermore, the incorporation of the FAFN further refines high-frequency detail reconstruction and improves the quality of fused images.
2) *Spatiotemporal Dependency-Aware Fusion Strategy:* Existing fusion algorithms often neglect explicit spatiotemporal-related modeling. Our two-stage fusion technique addresses this by decoupling temporal and spatial attention mechanisms, resulting in improved performance and remarkable interpretability, a significant achievement for real-world remote sensing applications that require traceable decision-making.
3) *Comprehensive Validation:* Extensive experiments on multiple STF datasets (e.g., LGC, Daxing, and AHB) show that our method outperforms SOTA techniques in terms of both quantitative measures and qualitative outcomes, indicating its broad application to remote sensing tasks.

The rest of this article is structured as follows. The related work is reviewed in Section II. Section III details the model architecture, including the agent attention mechanism and fusion strategy. Section IV presents the experimental setup and results, and Section V concludes this article with suggestions for future work.

II. Related Work

Over the past few decades, numerous STF methods have been developed [2], [21], which can be broadly categorized into five classes [16]: weight function-based, unmixing-based, Bayesian-based, hybrid, and learning-based.

Weight Function-Based Methods: Utilize weight functions to measure similarities between coarse and fine images, enabling pixel-level fusion. The spatial and temporal adaptive reflectance fusion model (STARFM) [22] uses spectral change analysis to predict each pixel. Despite the introduction of advanced techniques with higher fusion accuracy and increased robustness [23], [24], the performance of weight function-based methods tends to decline in regions with significant heterogeneity and complicated fluctuations in land cover.

Unmixing-Based Methods: Use linear spectral mixing theory to rebuild fine pictures by deconstructing spectral information from mixed pixels. The first method based on unmixing is multiscale mapping and transformation (MMT) [25] and many similar methods followed, including the spatiotemporal data fusion approach (STDFA) [26]. Although the performance of these methods has been improved, the assumption of linear interactions among pixels may not always hold true in real-world scenarios.

Bayesian-Based Methods: Treat STF as a probabilistic estimation problem, using maximum a posteriori (MAP) estimation to generate fusion results. Current Bayesian-based methods [27], [28] create temporal and scale models to forecast the relationship between the input and predicted images. These techniques offer great flexibility and robustness, particularly in scenarios involving heterogeneous regions or where additional data is readily available.

Hybrid Methods: Have been developed to integrate weight functions, spectral unmixing, Bayesian estimation, and machine learning algorithms. For instance, the flexible spatiotemporal data fusion (FSDAF) [29] incorporates spectral linear unmixing theory with the thin plate spline interpolation method. These hybrid approaches [30], [31] not only facilitate the prediction of gradual land-cover transitions but also improve fusion accuracy by adapting to both steady and dynamic land-cover changes.

Recent advances in learning-based methods have significantly advanced STF through distinct approaches. CNN-based frameworks like StfNet [32] and EDCSTFN [17] established the foundation for end-to-end fusion, effectively capturing local spatial patterns through their hierarchical architectures. However, their limited receptive fields constrain long-range dependency modeling, particularly problematic for large-area land-cover changes. GAN-based methods, such as GAN-STFM [33] and CycleGAN-STF [34], introduced adversarial

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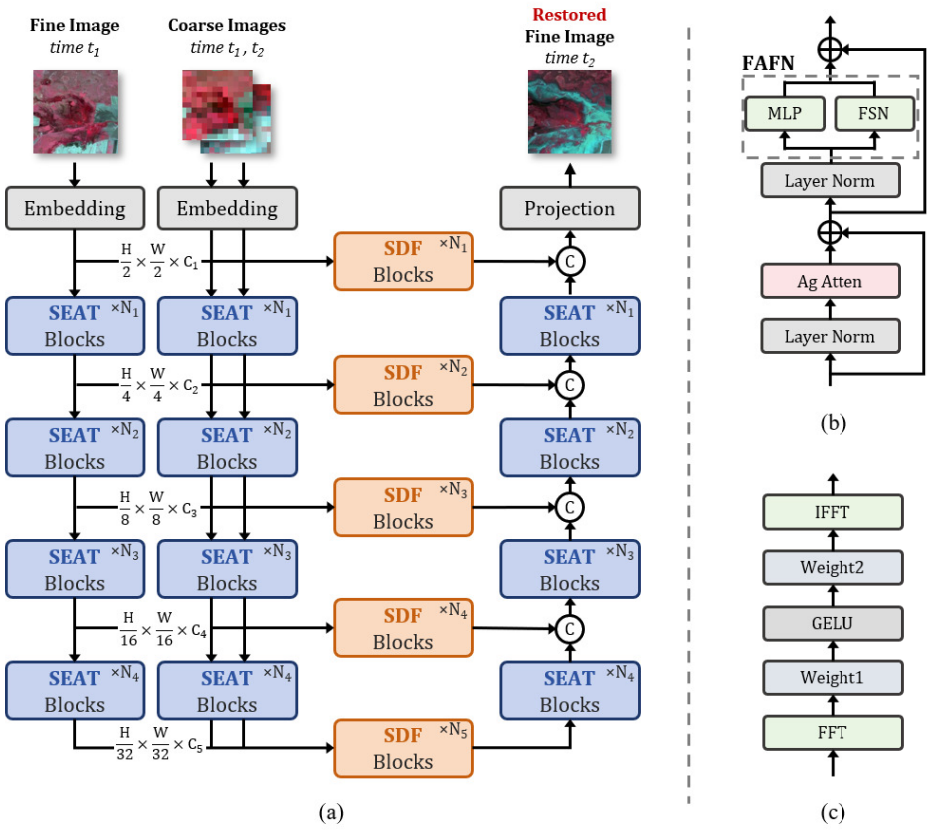


Fig. 1. Flowchart of the proposed FSDFormer. (a) Overview of the FSDFormer structure, which mainly consists of the SEAT block and the SDF block. (b) SEAT block, which comprises agent attention (Ag Atten) and FAFN. (c) FSN in FAFN, which utilizes FFT and iFFT for frequency-domain transformations.

training to enhance texture synthesis, demonstrating remarkable success in generating visually realistic high-resolution outputs. Yet, they remain prone to training instability and mode collapse, often producing artifacts in heterogeneous landscapes with complex temporal variations. Diffusion models, such as STFDiff [35], represent the most recent progress, applying iterative denoising to capture complicated spatiotemporal patterns, but their computational intensity and slow inference speed pose practical deployment challenges.

Furthermore, Transformer-based architectures have emerged as a promising alternative, resolving these constraints via global self-attention processes. MSNet [36] and SwinSTFM [37] excel at modeling spatiotemporal relationships over distant pixels, making them ideal for large-area remote sensing. However, current implementations face three major challenges: 1) quadratic computational complexity in connection to image size, limiting their scalability; 2) insufficient preservation of high-frequency details due to over-smoothing in attention aggregation; and 3) sensitivity to temporal misalignments in multidate observations. While SwinSTFM mitigates complexity through windowed attention, it sacrifices global context in favor of computational efficiency—a tradeoff that our work specifically addresses through novel architectural innovations.

III. Method

A. Overall Architecture

The proposed FSDFormer, as illustrated in Fig. 1(a), adopts a U-shaped hierarchical architecture with skip connections incorporating the encoder and decoder stages. The model uses a dataset comprising a coarse–fine image pair at the reference date (C_{t_1}, F_{t_1}) and a subsequent coarse image (C_{t_2}) at the target date to predict the fine image (F_{t_2}) at the target date.

The initial feature extraction involves a 2×2 convolution and layer normalization to process low-level features. Subsequently, FSDFormer employs four encoder stages composed of the spatial enhanced agent Transformer (SEAT) blocks to extract features from the input images, respectively. SEAT blocks capture long-range dependencies through self-attention, while the agent attention mechanism mitigates the computational demands of traditional Transformers, thereby balancing efficiency with global context modeling.

The output of the encoder is processed by the spatiotemporal differential fusion (SDF) block for multitemporal feature integration, enhancing the temporal coherence between images from different dates. The decoder mirrors the encoder's structure, using sequential SEAT blocks to refine the fused features and restore spatial details. Finally, the final projection module, which includes pixel shuffle, convolution, and activation functions, upscales the decoder output to produce the fine image F_{t_2}.

In addition, the symbols used in this article are defined in Table I.

B. SEAT Block

1) Revisiting Traditional Self-Attention: The Transformer model has demonstrated notable success across various

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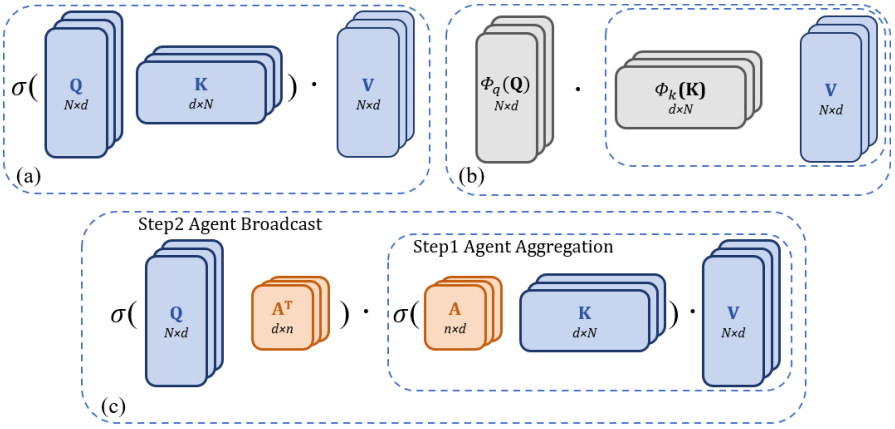


Fig. 2. Difference between the (a) Softmax attention, (b) linear attention, and (c) agent attention.

TABLE I
DEFINITIONS OF SYMBOLS USED IN THIS ARTICLE

Symbol	Definition
C_{t_1}, C_{t_2}	Coarse images at the reference date and predicted date, respectively.
F_{t_1}, F_{t_2}	Fine images at the reference date and predicted date, respectively.
$\Delta C, \Delta F$	Difference between reference date and predicted date of coarse and fine images, respectively.
C, H, W	Number of channels, height, and width of the features.
X_l	The feature maps produced by the l-th stage of blocks.

computer vision tasks [38], including semantic segmentation [39], [40], [41], object detection [42], [43], and image classification [44], [45]. This success is primarily attributed to the self-attention mechanism, which captures long-range dependencies by computing the pairwise similarity between all input tokens. As illustrated in Fig. 2(a), with an input of N tokens and C channels represented as $X \in \mathbb{R}^{N\times C}$, the traditional single-head Softmax attention mechanism [46] is defined as

$$Q = XW_Q, K = XW_K, V = XW_v \tag{1}$$

$$O_S = \sigma\left(\frac{QK^T}{\sqrt{d}}\right)V \tag{2}$$

where $Q, K, V \in \mathbb{R}^{N\times d}$ corresponded to the query, key, and value matrices, respectively; $W_{Q/K/V} \in \mathbb{R}^{C\times d}$ are projection matrices, while d is the head dimension. This formulation scales each similarity score to prevent gradient vanishing, while Softmax normalization (σ) amplifies critical features. Although effective in capturing dependencies, this mechanism results in $\mathcal{O}(N^2)$ computational complexity, with N representing the input length, making it computationally prohibitive for high-resolution images due to the attention matrix of size $N \times N$.

Linear attention [47] has been introduced as an alternative, using mapping functions $\phi_q()$ and $\phi_k()$ separately to Q and K to restructure the computation order, thereby reducing complexity to $\mathcal{O}(N)$ through leveraging the associative property of matrix multiplication, as depicted in Fig. 2(b). However, these methods often compromise performance for efficiency, as simpler mapping functions (e.g., ReLU) lose representational power, while more complex functions increase computational overhead. Consequently, traditional self-attention mechanisms, whether Softmax or linear, face a tradeoff between computational efficiency and model performance, particularly in high-resolution image tasks.

2) Mechanisms of the SEAT Block: To address these limitations, we propose the SEAT block, as depicted in Fig. 1(b). The SEAT block utilizes an agent attention mechanism [48] to minimize computational complexity while maintaining long-range dependencies. Furthermore, the block incorporates the FAFN, which employs fast Fourier transform operations to enhance local feature extraction. Specifically, given the input features X_{l-1} from the previous layer, the SEAT block comprises two key components: 1) agent attention (Ag Atten) and 2) FAFN. The computations of the SEAT block are described as

$$X'_l = \text{Ag Atten}(\text{LN}(X_{l-1})) + X_{l-1} \tag{3}$$

$$X_l = \text{FAFN}(\text{LN}(X'_l)) + X'_l \tag{4}$$

where X'_l and X_l represent outputs of the Ag Atten and FAFN modules, respectively, and LN denotes layer normalization. In the following, we elaborate on Ag Atten and FAFN separately.

3) Agent Attention (Ag Atten): The agent attention mechanism addresses the limitations of traditional Softmax and linear attention by introducing intermediary variables, termed agent tokens A, which enable efficient information aggregation and distribution between the key–value pairs (K, V) and query tokens Q. As depicted in Fig. 2(c), this mechanism allows agent tokens A to extract essential information from K and V, propagating it back to Q to achieve a balance between reduced computational load and rich contextual representation. To enhance positional sensitivity, an agent bias term is introduced, enabling each agent vector to concentrate on specific areas of the inputs, thereby improving spatial localization.

To address the issue of reduced feature diversity due to the diminished rank of the attention matrix in agent attention, we introduce a rank restoration module after the aggregation and broadcasting phases. This module employs an additional depthwise separable convolution (DWC) applied to the original attention matrix. Unlike standard convolution, which

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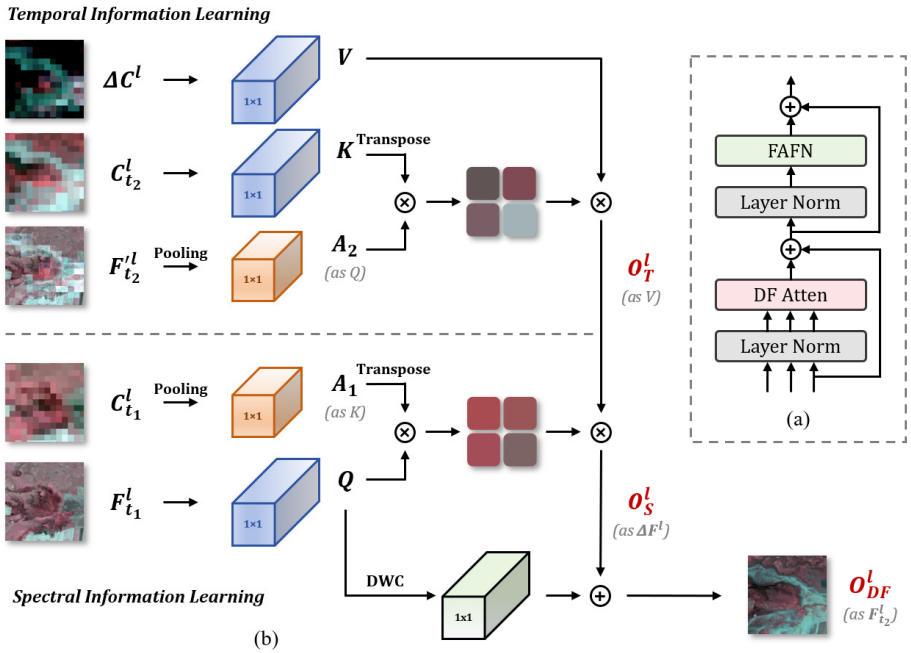


Fig. 3. Detail of the (a) SDF block, (b) differential fusion attention (DF Atten) in SDF, which is made up of temporal information learning and spatial information learning.

homogenizes features across the entire map, DWC preserves spatial uniqueness by applying distinct filters to different locations. This mechanism can be viewed as a specialized attention form, where each query focuses only on local features [49]. Consequently, this approach restores the matrix rank, diversifies positional output features, and achieves performance comparable to Softmax attention while maintaining low computational complexity. The complete Ag Atten operation is formulated as

$$O_A = \sigma(QA^T + B_1)\sigma(AK^T + B_2)V + \mathrm{DWC}(V) \tag{5}$$

where Q, K, and $V \in \mathbb{R}^{N\times d}$, and positional bias terms $B_1 \in \mathbb{R}^{n\times N}$ and $B_2 \in \mathbb{R}^{N\times n}$ enhance the spatial localization of agent tokens. The agent tokens A are derived from an adaptive average pooling operation on Q, specifically $A = \mathrm{Pooling}(Q) \in \mathbb{R}^{n\times d}$, which aggregates local features into representative values while preserving important contextual information. This pooling method strikes a balance between computational efficiency and feature representation richness, simplifying computation while retaining global contextual information.

The computational complexity of this mechanism is expressed as

$$\Omega = \underbrace{4NCd}_{\text{Proj}} + \underbrace{Nd}_{\text{Pooling}} + \underbrace{2nNd + 2Nnd}_{\text{Ag Atten}} + \underbrace{k^2Nd}_{\text{DWC}} \tag{6}$$

where n represents the number of agent tokens and $k = 3$ is the kernel size of the DWC. By ensuring that $n \ll N$, agent attention achieves an overall complexity of $\mathcal{O}(N)$, making it well-suited for processing high-resolution data and effectively capturing multiscale information across the input.

4) Frequency-Adaptive Filtering Network: The FAFN enhances the SEAT block by selectively amplifying informative frequency bands, thereby refining image representations for applications such as reconstruction and synthesis. As shown in Fig. 1(b), the FAFN comprises a multilayer perceptron (MLP) and a frequency selection network (FSN). The FSN, which performs adaptive filtering in the Fourier domain, is depicted in Fig. 1(c) and proceeds as follows:

$$\mathrm{FSN}(\mathrm{LN}(X'_l)) = \mathcal{F}^{-1}(\Phi(\mathcal{F}(\mathrm{LN}(X'_l)) \odot W_1) \odot W_2) \tag{7}$$

where X'_l is the output of Ag Atten, $\mathcal{F}$ and $\mathcal{F}^{-1}$ denote Fourier and inverse Fourier transforms, respectively, Φ is the activation function, and W_1 and W_2 are learned weights. The Fourier transform and its inverse are applied individually to each spectral band, allowing adaptive adjustment of frequency components within each band. This band-wise processing ensures that the inverse Fourier transform produces real-valued outputs by leveraging the inherent properties of the Fourier transform and its inverse, which maintain conjugate symmetry in the frequency domain. Consequently, valid image representations are preserved without requiring explicit real part extraction.

Overall, the FAFN enhances the SEAT block's capacity to manage diverse remote sensing datasets, improving the model's generalization and adaptability in addressing complex spatiotemporal challenges.

C. SDF Block

As depicted in Fig. 3(a), the SDF block differs solely in its self-attention computations compared to the SEAT block, specifically employing differential fusion attention (DF Atten) instead of Ag Atten. The subsequent discussion provide a detailed introduction to DF Atten.

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1) Overview of Fusion Methods in STF: Fusion methods in STF are broadly classified as increment- and regression-based approaches, which can be unified under a shared theoretical framework.

Increment-based methods estimate temporal changes between two fine images F_{t_1} and F_{t_2}, denoted as ΔF, which are subsequently added to F_{t_1} to generate the target fused image $\hat{F}_{t_2}$. This is expressed as

$$\Delta F = f(\Delta C) + \xi, \hat{F}_{t_2} = F_{t_1} + \Delta F \tag{8}$$

where ΔC represents observed changes in coarse images C_{t_1} and C_{t_2} and ξ encapsulates uncertainties stemming from observational or predictive inaccuracies.

Regression-based methods, on the other hand, directly map observations between the reference and target time, expressed as

$$C_{t_2} = a_c C_{t_1} + b_c, \quad \hat{F}_{t_2} = a_f F_{t_1} + b_f \tag{9}$$

where $a_{c/f}$ and $b_{c/f}$ are coefficients learned from coarse and fine-resolution data, respectively. Here, $a_{c/f}$ denotes scaling factors, while $b_{c/f}$ accounts for biases induced by temporal dynamics.

DF Atten adopts an increment-based strategy, evident in its core approach of predicting fine-grained changes (i.e., ΔF) and integrating them with the lth-layer feature representation of F_{t_1} (i.e., $F_{t_1}^l$) to generate the target fused output. This approach effectively captures subtle temporal variations between coarse and fine image representations.

2) Temporal Information Learning: DF Atten introduces a two-stage agent attention mechanism, with the first stage enhancing temporal dependencies using the temporal agent token A_2^l as the query. This token is derived from pooled features of $C_{t_2}^l$, directing the model's focus to significant temporal transitions. The temporal attention mechanism is formulated as

$$O_T^l = \sigma\left(\frac{A_2^l (K^l)^T}{\sqrt{d}}\right) V^l \tag{10}$$

where K^l and V^l represent key and value matrices computed from $\hat{F}_{t_2}^{\prime l} = F_{t_1}^l + \Delta C^l$ and ΔC^l, respectively. By aligning these matrices, DF Atten effectively captures critical temporal changes, enhancing its robustness in applications such as crop growth monitoring and other tasks involving phenological dynamics.

3) Spatial Information Learning: While capturing temporal dependencies is critical, maintaining spatial consistency is equally important, particularly in environments with complex spatial and phenological dynamics. To address this, DF Atten employs a second-stage attention mechanism that emphasizes spatial relationships. The spatial agent token A_1^l, derived by pooling features from $C_{t_1}^l$, guides spatial attention across feature representations as the key, while the output O_T^l from temporal information learning acts as the value. The spatial attention mechanism is defined as

$$O_S^l = \sigma\left(\frac{Q^l (A_1^l)^T}{\sqrt{d}}\right) O_T^l \tag{11}$$

where Q^l is derived from $F_{t_1}^l$. This two-stage attention mechanism enables DF Atten to effectively integrate spatial context with temporal dynamics, thereby enhancing its capability to capture complex spatiotemporal interactions.

TABLE II
SPECIFIC DATES FOR THE DAXING DATASET, AHB DATASET, AND LGC DATASET

Dataset	Daxing		AHB		LGC	
	Ref	Pred	Ref	Pred	Ref	Pred
Train	20130901	20131104	20130530	20140210	20040416	20040502
	20140429	20140819	20140415	20140906	20040502	20040605
	20140904	20141006	20150213	20150317	20040605	20040806
	20141225	20150211	20150504	20150621	20040806	20040822
	20150315	20150416	20150707	20150925	20040822	20041025
	20150518	20150822	20150925	20160131	20050113	20050129
	20170710	20170912	20170525	20170610	20050129	20050214
	20180203	20180408	20170728	20170829	20050214	20050302
	20181001	20181204	20180104	20180205	20050302	20050403
	20190121	20190529	20180325	20180512		
	20190614	20190817	20181003	20181019		
	20190918	20191105	20181019	20181206		
Test	20160214	20161214	20160420	20160607	20041126	20041212
	20161214	20170507	20160826	20161013	20041212	20041228
	20171030	20171115	20161013	20170218		

4) Differential Fusion Attention: To further enhance feature representation and diversity, a DWC is applied to Q before the attention computations. This operation preserves unique feature characteristics while maintaining computational efficiency. The outputs of the two attention stages are combined to generate a unified representation that encapsulates both temporal and spatial information. The final output of DF Atten, O_{DF}^l, is defined as shown in Fig. 3(b)

$$\begin{aligned} Q^l &= F_{t_1}^l W_Q^l, K^l = \hat{F}_{t_2}^{\prime l} W_K^l, V = \Delta C^l W_V^l \\ A_1^l &= \mathrm{Pooling}(C_{t_1}^l W_{A_1}^l), A_2^l = \mathrm{Pooling}(C_{t_2}^l W_{A_2}^l) \\ O_{\mathrm{DF}}^l &= \sigma(Q^l (A_1^l)^T + B_1^l)\sigma(A_2^l (K^l)^T + B_2^l) V^l + \mathrm{DWC}(Q^l) \end{aligned} \tag{12}$$

where $W_{Q/K/V/A_1/A_2}^l \in \mathbb{R}^{C\times d}$ are the projection matrices for the lth layer and B_1^l and B_2^l are positional biases introduced by the agent attention mechanism. This design leverages agent tokens A_1^l and A_2^l to integrate multiscale features effectively, enhances feature diversity through DWC, and ensures computational efficiency with linear complexity.

D. Loss Functions

We design a comprehensive loss function that combines three complementary components to enhance the performance of the fusion outcomes: gradient loss, perceptual loss, and frequency focus loss.

1) Gradient Loss: It mitigates outliers' impact while ensuring smooth gradient updates, which is crucial for image synthesis tasks. It is defined as

$$L_G = \frac{1}{N}\sum_{i=1}^{n} \sqrt{\left(\hat{F}_{t_2} - F_{t_2}\right)^2} + \epsilon^2 \tag{13}$$

where $\hat{F}_{t_2}$ and F_{t_2} represent the predicted and target images, respectively, with a small constant ϵ included to ensure computational stability. This loss minimizes pixel-wise reconstruction errors in a stable and outlier-resistant manner.

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TABLE III
PERFORMANCE ANALYSIS ON THE DAXING DATASET: BOLD VALUES REPRESENT THE BEST RESULTS, WHILE UNDERLINED VALUES INDICATE THE SECOND-BEST

	Band	STARFM	Fit-FC	OBSUM	FSDAF	ECPW	EDCSTFN	SwinSTFM	STFDiff	FSDFormer
RMSE	Blue	0.0128	0.0216	0.0109	0.0123	0.0107	0.0106	0.0235	0.0186	**0.0086**
	Green	0.0153	0.0219	0.0136	0.0143	0.0135	0.0129	0.0272	0.0277	**0.0111**
	Red	0.0192	0.0195	0.0168	0.0183	0.0169	0.0165	0.0227	0.0302	**0.0145**
	NIR	0.0302	0.0317	0.0319	0.0298	0.0342	0.0302	0.0382	0.0395	**0.0245**
	Average	0.0194	0.0237	0.0183	0.0187	0.0188	0.0175	0.0210	0.0290	**0.0147**
SSIM	Blue	0.8756	0.8026	0.9175	0.9030	0.9088	0.9134	0.7717	0.6690	**0.9427**
	Green	0.8494	0.8090	0.8914	0.8796	0.8856	0.8959	0.7861	0.5983	**0.9191**
	Red	0.7863	0.8075	0.8516	0.8300	0.8304	0.8459	0.7898	0.6041	**0.8785**
	NIR	0.6923	0.6742	0.6812	0.7021	0.6999	0.7337	0.6766	0.6267	**0.7655**
	Average	0.8009	0.7733	0.8354	0.8287	0.8312	0.8472	0.8036	0.6245	**0.8764**
CC	Blue	0.6515	0.5518	0.7732	0.7096	0.7815	0.7985	0.7514	0.6133	**0.8627**
	Green	0.6023	0.5732	0.7145	0.6823	0.7369	0.7470	0.7138	0.5379	**0.8212**
	Red	0.5872	0.6285	0.7120	0.6488	0.7129	0.7258	0.6918	0.5637	**0.8010**
	NIR	0.6496	0.6216	0.6320	0.6671	0.6550	0.6897	0.6441	0.6540	**0.7763**
	Average	0.6227	0.5938	0.7079	0.6769	0.7215	0.7403	0.6912	0.5922	**0.8153**
ERGAS		1.7510	2.4672	1.5364	1.6653	1.5357	1.4847	1.5487	2.2499	**1.2569**
SAM		17.9637	22.7578	15.9609	17.1178	15.6162	15.0369	17.3200	26.9842	**12.9107**

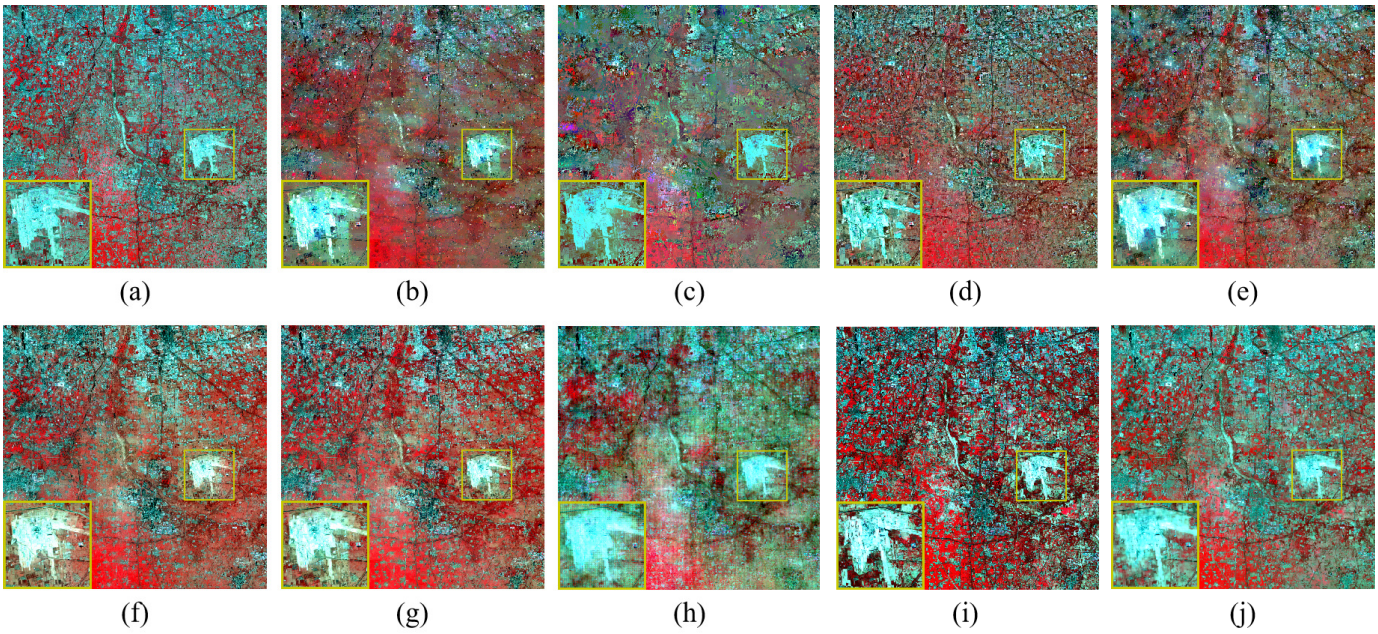


Fig. 4. Fusion results on the Daxing dataset. (a) GT. (b) STARFM. (c) Fit-FC. (d) OBSUM. (e) FSDAF. (f) ECPW. (g) EDCSTFN. (h) SwinSTFM. (i) STFDiff. (j) FSDFormer.

2) Perceptual Loss: It is derived from the multiscale structural similarity index (MS-SSIM), which quantifies structural similarity between predicted and target images. The loss function is defined as

$$L_P = 1 - \text{MS-SSIM}(\hat{F}_{t_2}, F_{t_2}) \tag{14}$$

where MS-SSIM computes structural similarity across multiple scales, providing a comprehensive assessment of image quality.

3) Frequency Focus Loss: It emphasizes challenging frequency components while down-weighting those that are easier to synthesize. An adaptive weighting matrix is constructed in this dynamic frequency-domain approach, with weights determined by the Euclidean distance comparing the frequency components of the target and predicted images. It is formulated as

$$L_F = \frac{1}{N} \sum_{u,v} w(u,v) \cdot \sqrt{\left|\hat{F}_{t_2}(u,v) - F_{t_2}(u,v)\right|^2} \tag{15}$$

where $\hat{F}_{t_2}(u,v)$ and $F_{t_2}(u,v)$ are the frequency components of the predicted and target images, respectively, $w(u,v)$ is a dynamic weight matrix (ranging from 0 to 1) that prioritizes regions with significant discrepancies. This ensures the model focuses on areas with substantial deviations, improving the reconstruction of fine details and achieving consistency in the frequency domain.

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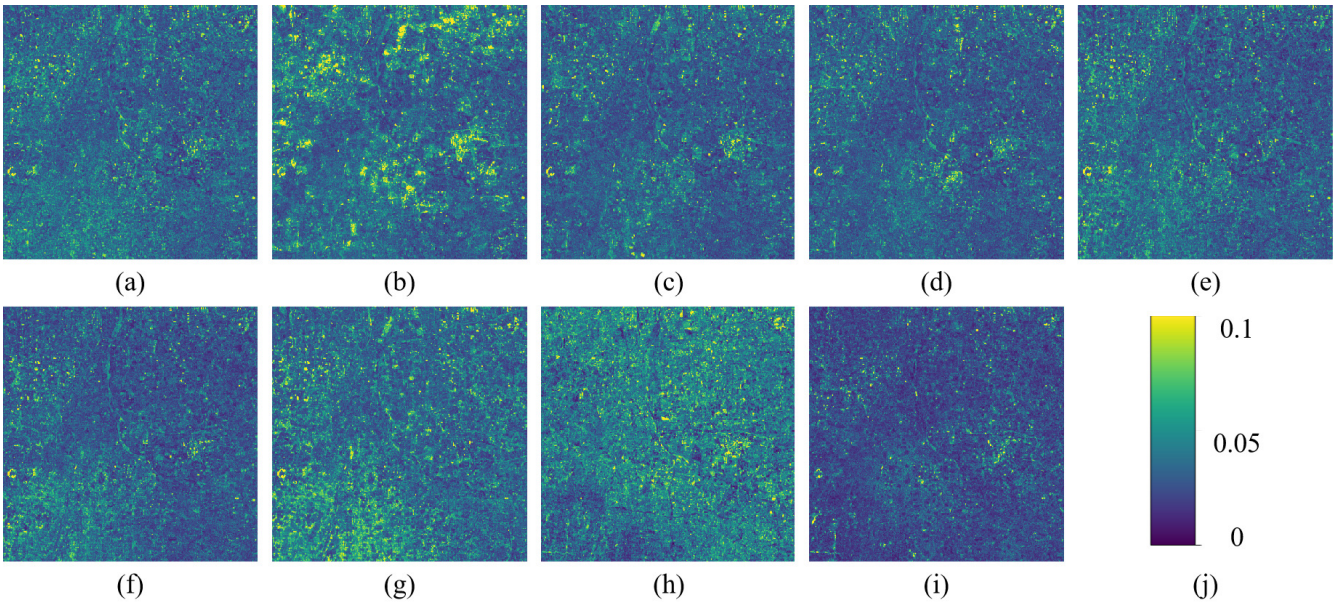


Fig. 5. AAD maps on the Daxing dataset. (a) STARFM. (b) Fit-FC. (c) OBSUM. (d) FSDAF. (e) ECPW. (f) EDCSTFN. (g) SwinSTFM. (h) STFDiff. (i) FSDFormer. (j) Color legend.

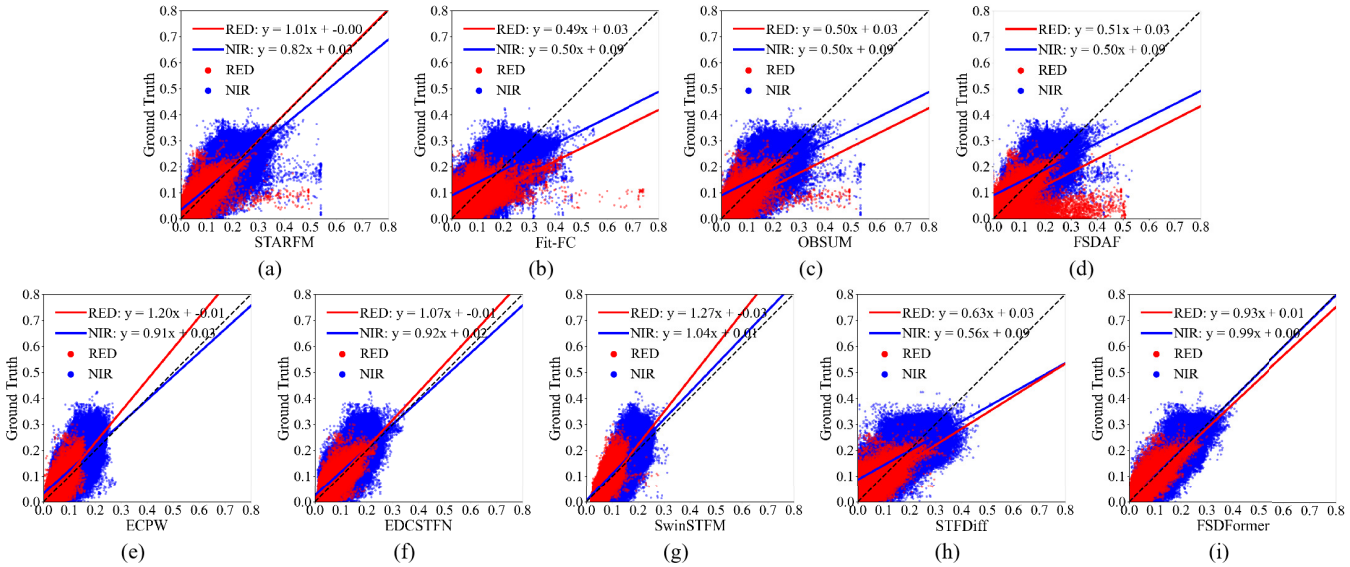


Fig. 6. Scatter diagrams of RED and NIR bands for the Daxing dataset. (a) STARFM. (b) Fit-FC. (c) OBSUM. (d) FSDAF. (e) ECPW. (f) EDCSTFN. (g) SwinSTFM. (h) STFDiff. (i) FSDFormer.

The total loss function L_{total} combines these elements as a weighted sum

$$L_{\text{total}} = L_G + \lambda_1 L_P + \lambda_2 L_F \tag{16}$$

where $\lambda_1 = 1$ and $\lambda_2 = 0.1$ regulate the contributions of perceptual loss and frequency focus loss, respectively. These values were chosen to balance the emphasis on structural similarity and fine-detail reconstruction effectively.

This composite loss function facilitates the generation of high-quality fused images that visually resemble the target images while exhibiting structural integrity and frequency domain consistency. By synergistically combining pixel-level, perceptual, and frequency-based constraints, the proposed approach achieves a well-rounded optimization that addresses the multifaceted demands of image fusion tasks.

IV. Experiment

A. Datasets and Implementation Details

This study employed three public datasets, each representing different geographical and environmental characteristics. The first dataset was collected from Daxing, Beijing, China (Daxing) [50], a rapidly urbanizing area with varied land-use types, including farmland, pastures, mountainous regions, and urban centers. The dataset spans from 2013 to 2019, encompassing 29 pairs of MODIS–Landsat images with an original resolution of 1640 × 1640 pixels across six bands. The second dataset originated from Ar Horqin Banner in

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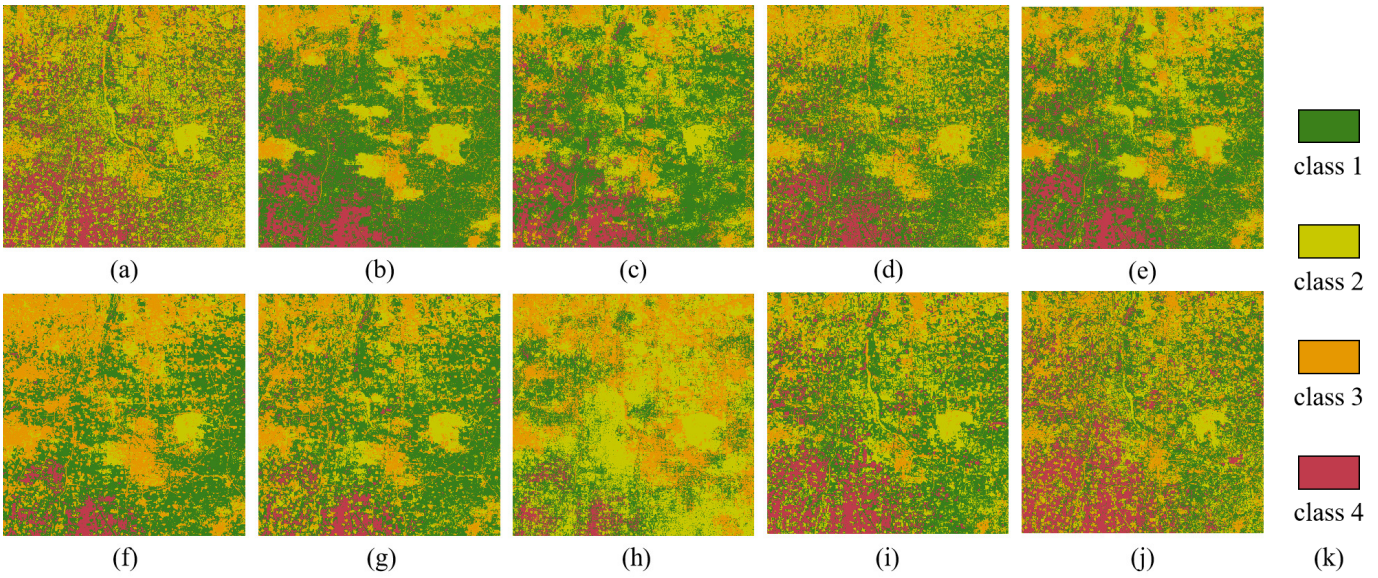


Fig. 7. *K*-means classification results on the Daxing dataset. (a) GT. (b) STARFM. (c) Fit-FC. (d) OBSUM. (e) FSDAF. (f) ECPW. (g) EDCSTFN. (h) SwinSTFM. (i) STFDiff. (j) FSDFormer. (k) Legend for classification outcomes.

TABLE IV
PERFORMANCE ANALYSIS ON THE AHB DATASET: BOLD VALUES REPRESENT THE BEST RESULTS, WHILE UNDERLINED VALUES INDICATE THE SECOND-BEST

	Band	STARFM	Fit-FC	OBSUM	FSDAF	ECPW	EDCSTFN	SwinSTFM	STFDiff	FSDFormer
RMSE	Blue	0.0109	0.0356	0.0302	0.0121	0.0039	0.0048	0.0066	0.0089	**0.0030**
	Green	0.0133	0.0352	0.0470	0.0150	0.0051	0.0085	0.0048	0.0055	**0.0034**
	Red	0.0192	0.0403	0.0588	0.0216	0.0098	0.0096	0.0080	0.0064	**0.0047**
	NIR	0.0173	0.0569	0.0788	0.0190	0.0175	0.0129	0.0140	0.0072	**0.0059**
	Average	0.0151	0.0420	0.0537	0.0169	0.0091	0.0089	0.0083	0.0070	**0.0042**
SSIM	Blue	0.6899	0.6012	0.6229	0.8703	0.9782	0.9634	0.9444	0.9440	**0.9889**
	Green	0.7376	0.6680	0.6286	0.9035	0.9790	0.9407	0.9840	0.9780	**0.9915**
	Red	0.7100	0.7162	0.6726	0.8941	0.9557	0.9622	0.9735	0.9805	**0.9883**
	NIR	0.9129	0.7854	0.7132	0.9165	0.9275	0.9631	0.9595	0.9804	**0.9832**
	Average	0.7626	0.6927	0.6593	0.8961	0.9601	0.9574	0.9653	0.9707	**0.9880**
CC	Blue	0.5654	0.3744	0.6960	0.6625	0.7765	0.7124	0.3599	0.2482	**0.7906**
	Green	0.5998	0.7487	0.7488	0.7459	0.8163	0.8926	0.6856	0.6969	**0.8964**
	Red	0.6784	0.7150	0.7569	0.7510	0.8265	0.9096	0.8223	0.8621	**0.9259**
	NIR	0.6392	0.4674	0.7481	0.7472	0.7688	**0.9240**	0.8589	0.8675	0.9131
	Average	0.6207	0.5764	0.7375	0.7267	0.7971	0.8596	0.6817	0.6687	**0.8815**
ERGAS		2.6831	7.6007	8.0028	2.8249	1.1765	1.3409	1.3392	1.2827	**0.6766**
SAM		26.6837	15.8569	9.1135	9.9021	8.1296	6.8654	10.0541	12.6570	**6.5919**

Inner Mongolia, China (AHB) [50], an area renowned for its rich agricultural landscape and pronounced seasonal variation. The dataset spans from 2013 to 2018, containing 27 MODIS–Landsat image pairs with a resolution of 2480×2800 pixels and six spectral bands. Located in New South Wales, Australia, the third dataset, Lower Gwydir catchment (LGC) [51] is a region known for its dynamic seasonal flooding and temporary land-cover changes. The time period extends from 2004 to 2005, consisting of 14 MODIS–Landsat image pairs at a resolution of 3200×2720 pixels across six bands.

Each dataset is divided into training and testing subsets with an approximately 4:1 ratio. The specific dates of the training datasets and testing datasets, along with the target images and reference images of the datasets, are shown in Table II.

During training, patches of size 256×256 were extracted from the images with a stride of 128. Testing sets followed a similar process, where the inputs were divided into patches, and the individual predictions were then merged to reconstruct the full-resolution output. The models were trained with the AdamW optimizer, starting with an initial learning rate of 0.0001 and following a polynomial decay schedule that included a warm-up phase. During the first 10 epochs (warm-up phase), the learning rate was gradually increased, after which it decayed according to a polynomial power of 0.9. Training was conducted for 200 epochs, with the batch size set to 8 and the final epoch determined by validation performance. All experiments were conducted on a machine equipped with an NVIDIA RTX 3090 GPU (24GB), a 14-core Intel Xeon Gold 6330 CPU (2.00 GHz), and Ubuntu 20.04.

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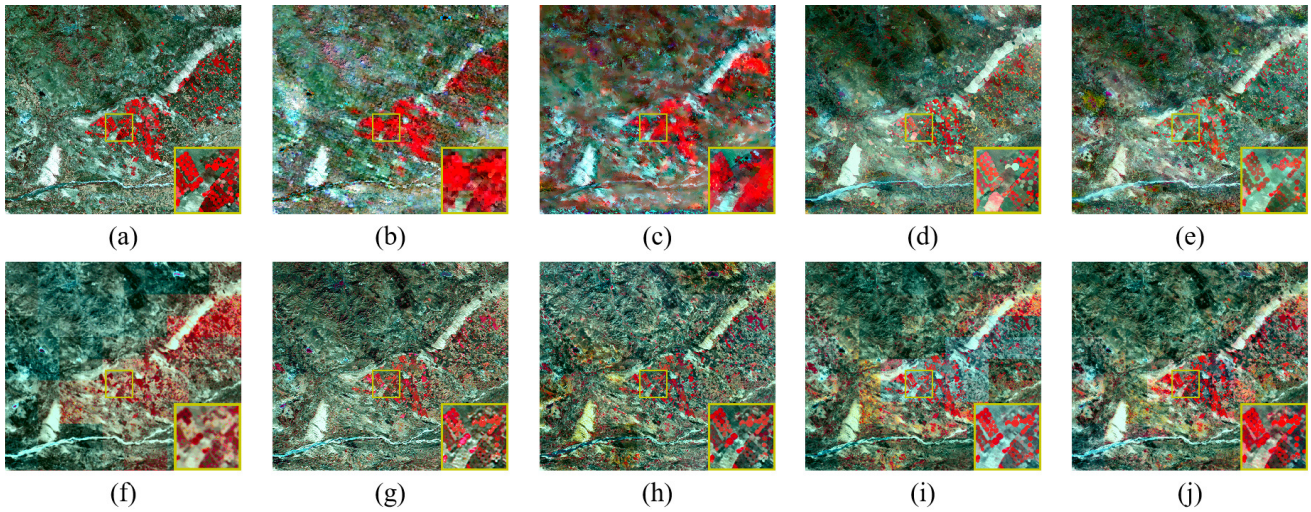


Fig. 8. Fusion results on the AHB dataset. (a) GT. (b) STARFM. (c) Fit-FC. (d) OBSUM. (e) FSDAF. (f) ECPW. (g) EDCSTFN. (h) SwinSTFM. (i) STFDiff. (j) FSDFormer.

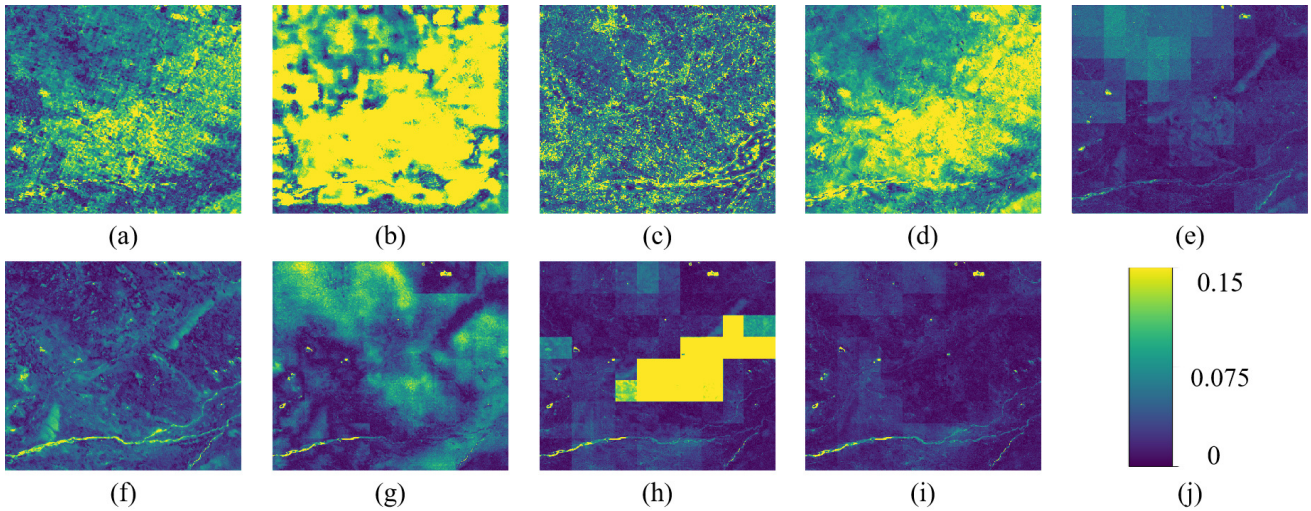


Fig. 9. AAD maps on the AHB dataset. (a) STARFM. (b) Fit-FC. (c) OBSUM. (d) FSDAF. (e) ECPW. (f) EDCSTFN. (g) SwinSTFM. (h) STFDiff. (i) FSDFormer. (j) Color legend.

The development environment included Python 3.8 and PyTorch 2.0.0.

B. Comparison Methods and Evaluation Indicators

For a comprehensive assessment, we conducted a comparative evaluation of FSDFormer with seven SOTA STF techniques: STARFM [22], Fit-FC [31], OBSUM [52], FSDAF [29], ECPW [53], EDCSTFN [17], SwinSTFM [37], and STFDiff [35]. These methods encompass a diverse range of strategies, spanning from traditional model-based approaches, hybrid frameworks, to CNN-based architectures, Transformer-based approaches, and diffusion models. This selection ensures a comprehensive comparison across multiple evaluation metrics, including precision, robustness, and generalization capabilities. To ensure consistency across all methods, similar settings were applied across all deep learning-based approaches. Traditional methods adhered to the parameter settings specified in their respective original papers.

Five key performance metrics were rigorously applied to evaluate the model's efficacy. The root-mean-square error (RMSE) quantifies the precision of spectral restoration, while the structural similarity index (SSIM) [54] assesses the integrity of spatial structure and the quality of spatial information. The correlation coefficient (CC) [55] measures the accuracy of spatial reconstruction. Furthermore, the spectral angle mapper (SAM) [56] evaluates the fidelity of spectral curve representation, and the erreur relative globale adimensionnelle de synthése (ERGAS) [57] measures the overall geometric and radiometric quality of the fusion process. Ideal values for these metrics are 0 for RMSE, SAM, and ERGAS, and 1 for SSIM and CC. The fusion results are presented in false color, utilizing the near-infrared, red, and green bands to enhance visualization.

C. Fusion Results on Daxing

Capturing urban development during the construction of the Beijing Daxing International Airport between December 2014

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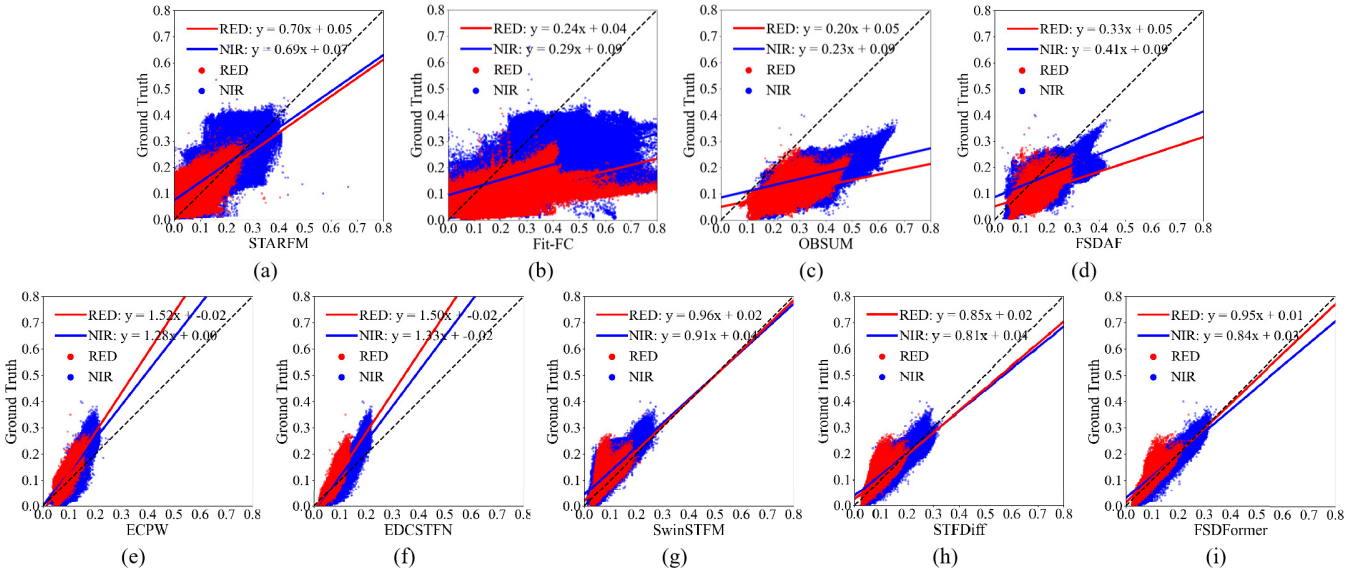


Fig. 10. Scatter diagrams of RED and NIR bands for the AHB dataset. (a) STARFM. (b) Fit-FC. (c) OBSUM. (d) FSDAF. (e) ECPW. (f) EDCSTFN. (g) SwinSTFM. (h) STFDiff. (i) FSDFormer.

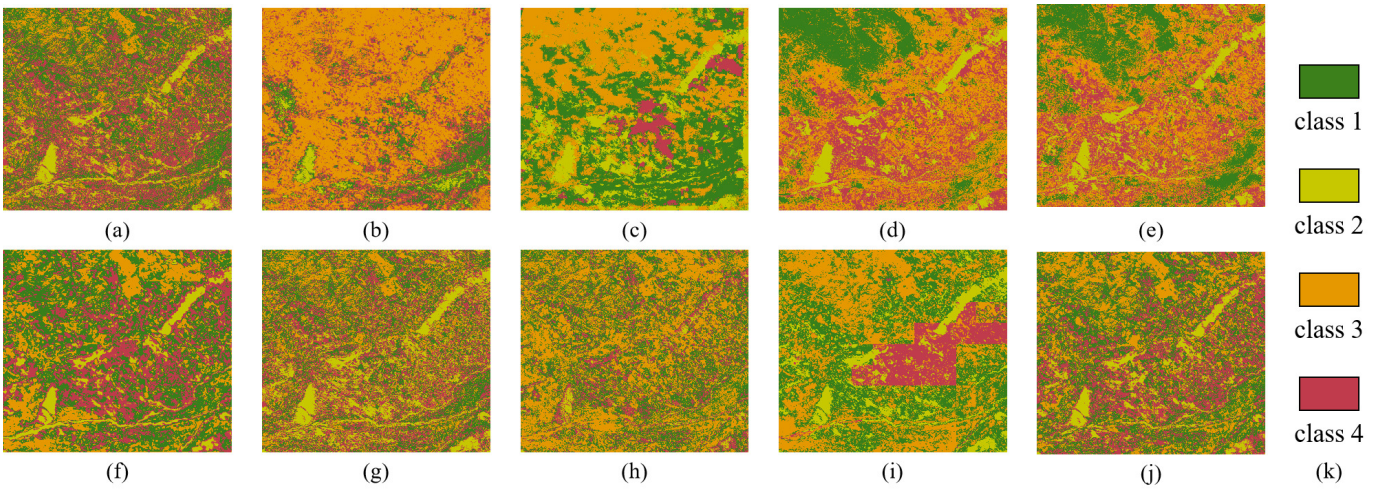


Fig. 11. *K*-means classification results on the AHB dataset. (a) GT. (b) STARFM. (c) Fit-FC. (d) OBSUM. (e) FSDAF. (f) ECPW. (g) EDCSTFN. (h) SwinSTFM. (i) STFDiff. (j) FSDFormer. (k) Legend for classification outcomes.

and September 2019, the Daxing dataset is widely recognized for its diversity and complexity. It encompasses a substantial temporal span, intricate land-cover changes, and fine-grained texture details, offering a robust testbed for evaluating fusion methods. To demonstrate these characteristics, we strategically chose test subsets containing pronounced phenological shifts and anthropogenic modifications, for example, December 14, 2016, to May 7, 2017, as shown in Fig. A1 in the Supplementary Material, which captures both winter–spring vegetation transitions and early-stage airport structural development.

To thoroughly evaluate the performance of FSDFormer, we initially present a quantitative comparison of fusion accuracy in Table III. The table outlines the performance of various methods across key evaluation metrics on the first four bands (blue, green, red, and NIR), which were retained due to stripe noise in bands 5 and 6. FSDFormer consistently outperforms all competitors across all four bands, achieving the highest average scores in all five metrics. Complementing this, Fig. 4 illustrates the results generated by eight STF methods for May 7, 2017, focusing on the subregion encompassing Daxing Airport. A visual inspection reveals that FSDFormer achieves the highest resemblance to ground truth, with superior recovery of texture details and minimal artifacts, underscoring its capacity to faithfully reconstruct spatial and temporal information.

Further analysis is facilitated by Fig. 5, which displays the absolute average deviation (AAD) map, a quantitative tool for identifying discrepancies between predictions and actual observations. Additionally, Fig. 6 offers scatter diagrams illustrating the correlation between predicted and actual values, where the compact clustering of points in the FSDFormer results reflects its strong alignment with the

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TABLE V

PERFORMANCE ANALYSIS ON THE LGC DATASET: BOLD VALUES REPRESENT THE BEST RESULTS, WHILE UNDERLINED VALUES INDICATE THE SECOND-BEST

	Band	STARFM	Fit-FC	OBSUM	FSDAF	ECPW	EDCSTFN	SwinSTFM	STFDiff	FSDFormer
RMSE	Blue	0.0154	0.0161	0.0178	0.0152	0.0178	0.0125	**0.0112**	0.0158	0.0114
	Green	0.0185	0.0171	0.0187	0.0199	0.0209	0.0165	**0.0128**	0.0183	0.0130
	Red	0.0245	0.0231	0.0237	0.0399	0.0245	0.0205	0.0158	0.0213	**0.0155**
	NIR	0.0368	0.0401	0.0374	0.0381	0.0463	0.0475	0.0324	0.0400	**0.0312**
	SWIR1	0.0509	0.0545	0.0534	0.0589	0.0500	0.0475	0.0370	0.0480	**0.0354**
	SWIR2	0.0442	0.0591	0.0446	0.0576	0.0427	0.0419	0.0331	0.0384	**0.0328**
	Average	0.0317	0.0350	0.0326	0.0383	0.0337	0.0310	0.0238	0.0303	**0.0232**
SSIM	Blue	0.9039	0.9016	0.8767	0.8967	0.9040	0.9269	**0.9458**	0.9145	0.9452
	Green	0.8837	0.8960	0.8765	0.8704	0.8915	0.9040	0.9333	0.9020	**0.9355**
	Red	0.8271	0.8429	0.8369	0.8081	0.8531	0.8622	0.9032	0.8606	**0.9086**
	NIR	0.7530	0.7519	0.7689	0.7449	0.7419	0.7240	0.7858	0.7076	**0.8089**
	SWIR1	0.6456	0.6266	0.6317	0.6316	0.6867	0.6913	0.7515	0.6796	**0.7693**
	SWIR2	0.6734	0.6146	0.6752	0.6487	0.7038	0.6967	0.7683	0.7068	**0.7831**
	Average	0.7811	0.7723	0.7776	0.7667	0.7968	0.8008	0.8480	0.7952	**0.8584**
CC	Blue	0.7107	0.7017	0.7228	0.7114	0.6893	0.7559	0.8491	0.7357	**0.8608**
	Green	0.7072	0.7366	0.7187	0.7150	0.6825	0.7559	0.8600	0.7703	**0.8636**
	Red	0.7382	0.7621	0.7636	0.5644	0.7731	0.7972	0.8809	0.8129	**0.8866**
	NIR	0.8646	0.8518	0.8576	0.8674	0.8177	0.8161	0.8934	0.8424	**0.9079**
	SWIR1	0.8139	0.7659	0.7769	0.7762	0.7802	0.7932	0.8804	0.8223	**0.8894**
	SWIR2	0.8114	0.7191	0.7933	0.7369	0.7951	0.8057	0.8828	0.8388	**0.8924**
	Average	0.7744	0.7562	0.7721	0.7286	0.7563	0.7873	0.8744	0.8073	**0.8835**
ERGAS		1.6053	1.7497	1.6736	2.0092	1.7105	1.4726	1.1558	1.5233	**1.1514**
SAM		13.8066	14.6592	14.2187	16.6293	13.4238	12.7284	9.9172	12.1825	**9.5836**

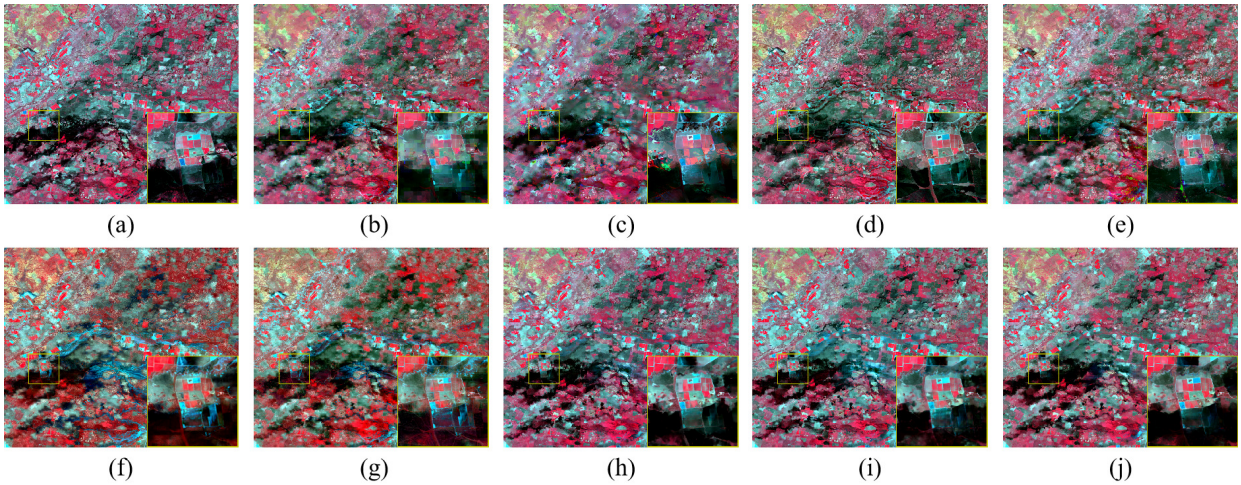


Fig. 12. Fusion results on the LGC dataset. (a) GT. (b) STARFM. (c) Fit-FC. (d) OBSUM. (e) FSDAF. (f) ECPW. (g) EDCSTFN. (h) SwinSTFM. (i) STFDiff. (j) FSDFormer.

TABLE VI

COMPUTATIONAL COMPLEXITY, FLOPS, AND PARAMETERS' COMPARISON FOR BACKBONE ABLATION STUDY

Backbone	Complexity	Flops	Params	FPS
ViT	$\mathcal{O}(N^2)$	23.34G	23.36M	4.09
Lin-T	$\mathcal{O}(N)$	23.23G	23.33M	12.15
ours	$\mathcal{O}(N)$	23.44G	23.37M	11.68

TABLE VII

ABLATION STUDY OF BACKBONE ON THREE DATASETS: BEST RESULTS ARE HIGHLIGHTED IN BOLD

Dataset	Backbone	RMSE	SSIM	CC	ERGAS	SAM
Daxing	**ViT**	0.0148	0.8756	0.8128	1.2908	12.9352
	Lin-T	0.0149	0.8754	0.8109	1.3026	12.9587
	ours	**0.0147**	**0.8764**	**0.8153**	**1.2569**	**12.9107**
AHB	**ViT**	0.0064	0.9802	0.8135	0.7290	8.3495
	Lin-T	0.0055	0.9866	0.8149	0.6884	6.9891
	ours	**0.0042**	**0.9880**	**0.8815**	**0.6766**	**6.5919**
LGC	**ViT**	0.0289	0.8467	0.8617	1.1973	9.8867
	Lin-T	0.0338	0.8109	0.7951	1.7076	12.6006
	ours	**0.0232**	**0.8584**	**0.8835**	**1.1514**	**9.5836**

ground truth and the highest coefficient of determination. Collectively, these results underscore the exceptional ability of FSDFormer to handle long temporal spans while preserving intricate spatial details.

An essential application of STF is generating high-quality remote sensing time series for vegetation monitoring. Fig. A4 in the Supplementary Material illustrates the ground normalized difference vegetation index (GNDVI) derived from the fused images, which can be leveraged for agricultural monitoring and management. By tracking crop growth and health,

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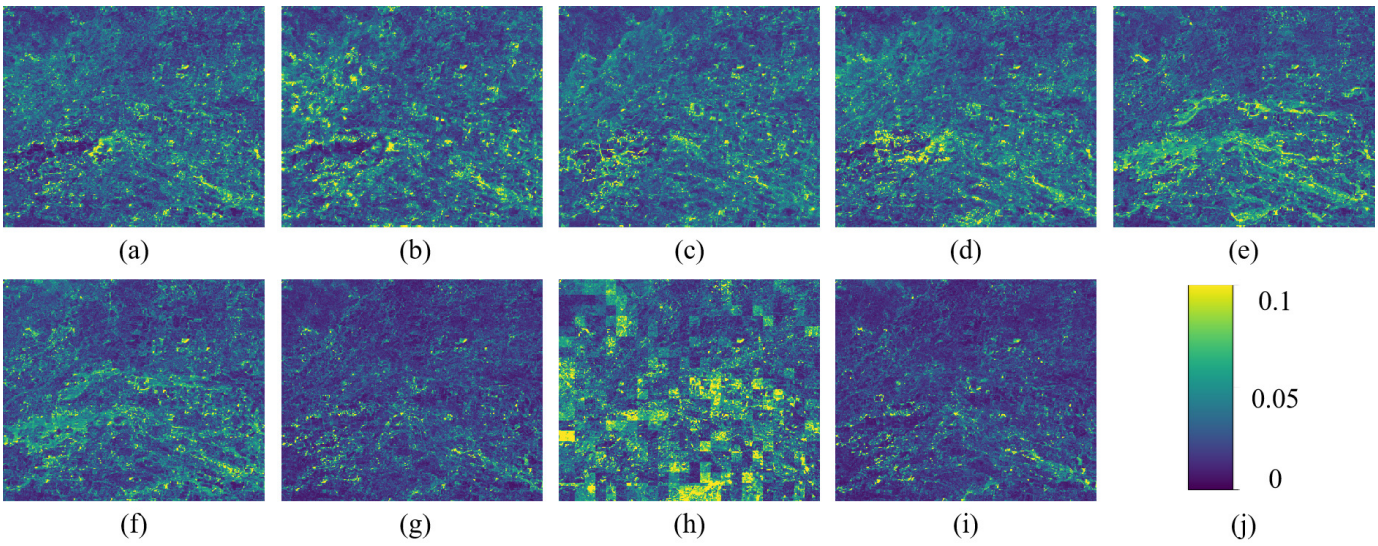


Fig. 13. AAD maps on the LGC dataset. (a) STARFM. (b) Fit-FC. (c) OBSUM. (d) FSDAF. (e) ECPW. (f) EDCSTFN. (g) SwinSTFM. (h) STFDiff. (i) FSDFormer. (j) Color legend.

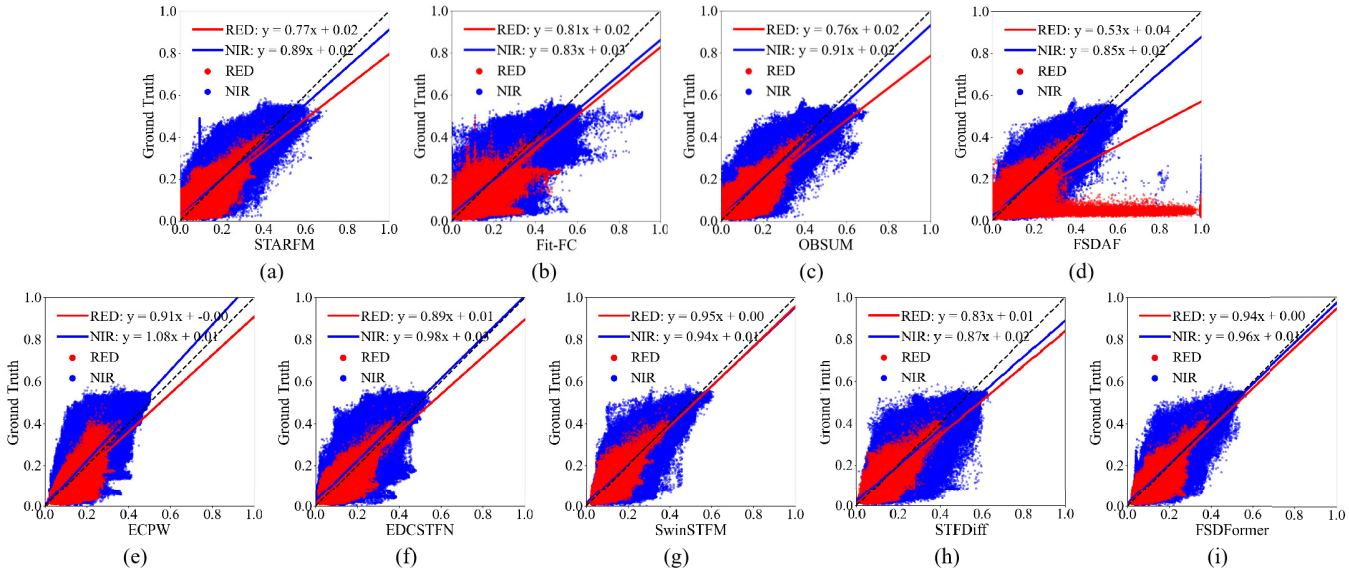


Fig. 14. Scatter diagrams of RED and NIR bands for the LGC dataset. (a) STARFM. (b) Fit-FC. (c) OBSUM. (d) FSDAF. (e) ECPW. (f) EDCSTFN. (g) SwinSTFM. (h) STFDiff. (i) FSDFormer.

the derived GNDVI series facilitates timely interventions to optimize yield and quality, demonstrating the practical utility of our method.

Furthermore, we employed the *K*-means clustering algorithm, a well-established technique in unsupervised data classification, which operates on the principle of iteratively optimizing cluster centroids to minimize the sum of squared errors within each cluster. To determine the optimal number of classes, we conducted a systematic evaluation of the dataset's inherent structure, ultimately identifying four clusters as the ideal tradeoff between classification granularity and computational tractability. Six metrics were used for quantitative validation: producer accuracy (PA), user accuracy (UA), intersection over union (IoU), mean IoU (mIoU), overall accuracy (OA), and Kappa coefficient. Fig. 7 and Table A1 in the Supplementary Material illustrate the clustering outcomes of our method, which exhibit a high degree of congruence with the ground truth. By comparing the discrepancies between the fusion outputs of various methods and the reference data, our approach demonstrated the lowest error across all categories. This not only highlights the advantages of our method in preserving spatial and temporal accuracy but also provides valuable data support for future applications.

D. Fusion Results on AHB

Spanning from May 30, 2013, to December 6, 2018, the AHB dataset documents phenological transitions in agricultural and mountainous regions, with notable temporal and spatial variations between observations, underscoring the

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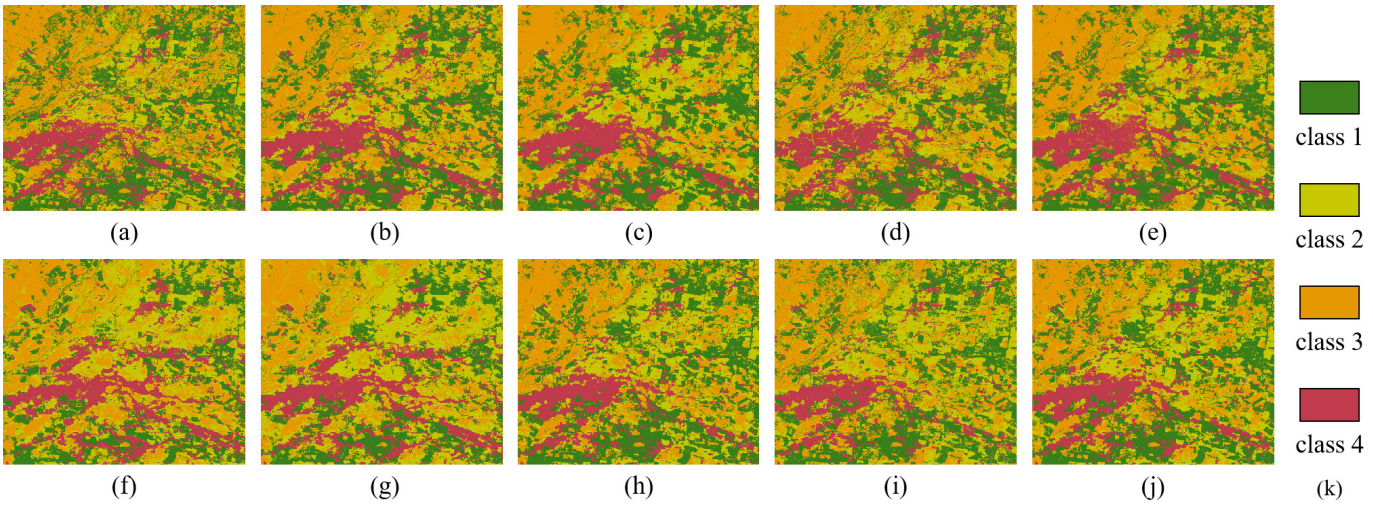


Fig. 15. *K*-means classification results on the LGC dataset. (a) GT. (b) STARFM. (c) Fit-FC. (d) OBSUM. (e) FSDAF. (f) ECPW. (g) EDCSTFN. (h) SwinSTFM. (i) STFDiff. (j) FSDFormer. (k) Legend for classification outcomes.

TABLE VIII

ABLATION STUDY OF THE FREQUENCY ENHANCEMENT NETWORK ON THREE DATASETS: BEST RESULTS ARE HIGHLIGHTED IN BOLD

Dataset	FFN	RMSE	SSIM	CC	ERGAS	SAM
Daxing	MLP	0.0148	0.8753	0.8118	1.2601	12.9832
	FSN	0.0152	0.8669	0.7966	1.3009	13.4339
	ours	**0.0147**	**0.8764**	**0.8153**	**1.2569**	**12.9107**
AHB	MLP	0.0046	0.9864	0.8011	0.7458	7.9256
	FSN	0.0067	0.9814	0.8729	0.7167	6.8344
	ours	**0.0042**	**0.9880**	**0.8815**	**0.6766**	**6.5919**
LGC	MLP	0.0235	0.8571	0.8819	1.1574	9.6642
	FSN	0.0236	0.8496	0.8738	1.1536	9.9362
	ours	**0.0232**	**0.8584**	**0.8835**	**1.1514**	**9.5836**

TABLE IX

ABLATION STUDY OF THE SDF NETWORK ON THREE DATASETS: BEST RESULTS ARE HIGHLIGHTED IN BOLD

Dataset	Fusion	RMSE	SSIM	CC	ERGAS	SAM
Daxing	Temporal	0.0149	0.8755	0.8144	1.3195	12.9815
	Spatial	0.0148	0.8757	0.8143	1.2857	12.9291
	Concat	0.0152	0.8741	0.8107	1.3303	13.0075
	Delta	0.0149	0.8754	0.8142	1.3184	12.9637
	ours	**0.0147**	**0.8764**	**0.8153**	**1.2569**	**12.9107**
AHB	Temporal	0.0051	0.9874	0.8739	0.8735	6.6744
	Spatial	0.0094	0.9569	0.7797	1.1648	8.5245
	Concat	0.0074	0.9612	0.6786	1.4767	10.1851
	Delta	0.0046	0.9877	0.8809	0.7242	7.0816
	ours	**0.0042**	**0.9880**	**0.8815**	**0.6766**	**6.5919**
LGC	Temporal	0.0234	0.8547	0.8722	1.1576	9.9687
	Spatial	0.0235	0.8556	0.8793	1.1531	9.7479
	Concat	0.0244	0.8482	0.8691	1.2435	10.0784
	Delta	0.0237	0.8563	0.8734	1.1739	9.9646
	ours	**0.0232**	**0.8584**	**0.8835**	**1.1514**	**9.5836**

inherent complexity of the dataset. The intricate land-cover changes and phenological variations present significant challenges for STF analysis. We adopted a similar testing dataset selection approach to that used for the Daxing dataset, choosing images with evident land-cover changes and phenological variations. Specifically, the pairs of August 26, 2016, to October 13, 2016 (Fig. A2 in the Supplementary Material) were selected to demonstrate the performance in the AHB dataset.

Fig. 8 presents a visual comparison of eight fusion methods for October 13, 2016, with a specific focus on their capacity to recover spatial details. FSDFormer stands out due to its ability to preserve texture and edge information, surpassing alternative methods in visual quality. Fig. 9 further supports this observation with the AAD maps, demonstrating the robustness of FSDFormer under significant phenological variations, where traditional methods, particularly Fit-FC, are less effective. Additionally, the scatter diagrams in Fig. 10 reaffirm these findings, as FSDFormer demonstrates the strongest correlation with actual data, reflected by its tight clustering and highest coefficient of determination.

Fig. A5 in the Supplementary Material and Fig. 11 provide further application-oriented analysis. Fig. A5 in the Supplementary Material presents GNDVI results derived from the fused images, while Fig. 11 and Table A2 in the Supplementary Material utilize *K*-means clustering to evaluate alignment with the ground truth. Both figures further substantiate FSDFormer's superior accuracy in capturing the intrinsic structure and fine-grained phenological details of the dataset.

Quantitative evaluations in Table IV validate these visual findings. FSDFormer achieves the highest scores across all four bands in nearly every metric, while exhibiting a negligible deviation in CC on the NIR band (ΔCC = 0.0109) compared to EDCSTFN. These results attest to FSDFormer's robustness and adaptability, confirming its dominance over both traditional and modern approaches.

In conclusion, FSDFormer excels in both qualitative and quantitative analyses on the AHB dataset, demonstrating its ability to address complex phenological dynamics and recover spatial details with precision. These results reinforce the method's potential for advancing STF research and its application in remote sensing.

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TABLE X
ABLATION STUDY OF THE LOSS FUNCTION ON THREE DATASETS: BEST RESULTS ARE HIGHLIGHTED IN BOLD

Dataset	L_G	L_P	L_F	RMSE	SSIM	CC	ERGAS	SAM
Daxing	✓			0.0172	0.8540	0.7862	1.2890	13.8056
		✓		0.0148	0.8755	0.8119	1.3259	12.9782
			✓	0.0154	0.8596	0.7872	1.3877	13.7303
	✓	✓		0.0148	0.8761	0.8139	1.2791	12.9913
		✓	✓	0.0148	0.8741	0.8092	1.3367	13.1054
	✓		✓	0.0150	0.8739	0.8073	1.3494	13.1989
	✓	✓	✓	**0.0147**	**0.8764**	**0.8153**	**1.2569**	**12.9107**
AHB	✓			0.0315	0.7791	0.5235	5.9792	11.4645
		✓		0.0107	0.9480	0.7283	1.5555	10.5477
			✓	0.0093	0.9370	0.5628	1.8158	11.8808
	✓	✓		0.0054	0.9773	0.8056	1.0892	7.7209
		✓	✓	0.0100	0.9502	0.7526	1.4254	8.5238
	✓		✓	0.0093	0.9482	0.6881	1.5580	10.2871
	✓	✓	✓	**0.0042**	**0.9880**	**0.8815**	**0.6766**	**6.5919**
LGC	✓			0.0242	0.8507	0.8635	1.1862	10.2724
		✓		0.0235	0.8528	0.8771	1.1519	9.8228
			✓	0.0241	0.8427	0.8708	1.1810	10.0542
	✓	✓		0.0233	0.8564	0.8790	1.1630	9.7433
		✓	✓	0.0235	0.8533	0.8771	1.1532	9.8229
	✓		✓	0.0234	0.8518	0.8766	1.1589	9.7987
	✓	✓	✓	**0.0232**	**0.8584**	**0.8835**	**1.1514**	**9.5836**

E. Fusion Results on LGC

The LGC dataset, spanning from April 16, 2004, to April 3, 2005, offers a distinctive benchmark for STF methods by capturing significant terrestrial changes in a relatively short timeframe. As a flood occurred in mid-December 2004, we selected the image pair of December 12 and 28, 2004 (Fig. A3 in the Supplementary Material), which brackets the peak flooding event and the subsequent water recession phase.

Table V presents the quantitative results for eight fusion methods. FSDFormer achieves the highest scores across most indicators, although SwinSTFM marginally surpasses it in a few metrics. The reduced temporal variability in the LGC dataset narrows the performance gap between traditional and deep learning approaches, making traditional methods more competitive regarding accuracy. Nonetheless, FSDFormer consistently exhibits superior robustness and generalizability when evaluated across multiple metrics.

The qualitative results in Fig. 12, corresponding to December 28, 2004, further highlight the differences. Both SwinSTFM and FSDFormer produce visually impressive results; however, FSDFormer exhibits fewer fusion artifacts and smoother edge transitions. This distinction is further emphasized in Figs. 13–15 as well as in Table A3 in the Supplementary Material and Fig. A6 in the Supplementary Material, which present AAD maps, scatter diagrams, clustering results, and GNDVI calculations. The enhanced naturalness and alignment with ground truth underscore the comprehensive advantages of FSDFormer in fusion tasks.

In summary, although the shorter temporal span of the LGC dataset somewhat diminishes the challenges of fusion, FSDFormer retains its advantage by delivering high-quality results across various metrics and visual analyses. This performance highlights its versatility and effectiveness, even under less challenging conditions, paving the way for broader applications.

F. Ablation Study

An in-depth ablation study is performed in this section to evaluate the contributions of the key components of FSDFormer across three datasets. These investigations are essential for elucidating the effectiveness of individual architectural elements and the loss function in enhancing the performance of our STF approach.

1) Backbone Ablation: To evaluate the influence of the backbone architecture, we compared our proposed backbone with two alternative designs: the traditional Vision Transformer (ViT) and the Linear Transformer (Lin-T). As shown in Table VI, despite differences in computational complexity, all alternatives were configured to ensure parity in FLOPs and parameters. Notably, our method achieves substantially higher frames per second (FPS) than the standard ViT while maintaining competitive FPS with the Lin-T variant. The results, presented in Table VII, show that our backbone consistently outperformed the ViT and Lin-T across all evaluation metrics. This superior performance highlights the efficacy of our backbone in capturing nuanced spatial hierarchies and long-range dependencies, which are crucial for handling the intricate characteristics of remote sensing data and achieving enhanced fusion performance.

2) Frequency-Adaptive Filtering Module Ablation: Next, we investigated the role of the FAFN, a critical component in the forward propagation mechanism of our model. To isolate its contributions, we conducted experiments using only the MLP or the FSN in isolation. The results, summarized in Table VIII, show that while each component offers certain benefits, the combined use of MLP and FSN in the FAFN framework yields the best performance. This synergy underscores the complementary roles of feature transformation and selective filtering in refining spatial and temporal details, thus enhancing the quality of the fused images.

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3) SDF Network Ablation: To further analyze the contributions of the SDF, we experimented with four configurations: 1) utilizing only the temporal information learning; 2) using only the spatial information learning; 3) directly concatenating features extracted from the encoder without SDF; and 4) replacing the SDF with the input $F_1' + (C_2' - C_1')$ for the decoder. Table IX summarizes the results, which consistently demonstrate that the complete SDF configuration outperforms all other variants. These findings highlight the critical role of integrating spatial and temporal information to achieve effective STF and reaffirm the necessity of the differential fusion strategy embedded in the SDF.

4) Loss Function Ablation: Finally, we examined the impact of our composite loss function, which integrates three distinct components: gradient loss (L_G), perception loss (L_P), and frequency focal loss (L_F). Through systematic ablation studies evaluating individual loss terms, pairwise combinations, and parameter sensitivity analysis [particularly for balancing coefficients λ_1 and λ_2 in (16)], we demonstrate the critical importance of proper component weighting. The comprehensive results (summarized in Tables X and A4 in the Supplementary Material) reveal that our full formulation with optimized coefficients achieves the best balance across content preservation, perceptual similarity, and frequency consistency. This balanced optimization is crucial for achieving high-quality STF outcomes, as it ensures robust reconstruction of both spatial details and temporal variations.

To evaluate the impact of the parameter ϵ in the gradient loss (13), we conducted an ablation study on the Daxing dataset (Table A5 in the Supplementary Material). Results demonstrate that $\epsilon = 0$ induces numerical instability, generating NaN (Not a Number) errors due to nonpositive definite matrices during square root computation. While larger values ($\epsilon \geq 10^{-1}$) marginally degrade accuracy, the influence becomes negligible at sufficiently small magnitudes ($\epsilon \leq 10^{-6}$). Consequently, we fix $\epsilon = 10^{-9}$ throughout this study, optimally balancing numerical stability with minimal precision tradeoffs.

Through these systematic ablation studies, we have demonstrated the essential contributions of each component in our model, from the backbone architecture to the FAFN, SDF, and loss function. These insights validate the design choices underlying our model and offer a deeper understanding of the mechanisms driving its superior performance in remote sensing image fusion tasks.

V. Conclusion

In this study, we introduced an innovative STF framework, FSDFormer, which integrates the specially designed SEAT, FAFN, and SDF, achieving substantial advancements in remote sensing image fusion. Notably, our fusion module, SDF, grounded in both advanced algorithmic design and a spatiotemporal dependency-aware fusion strategy, enhances fusion performance through two sequential stages of temporal and spatial information enhancement, delivering efficient and high-quality results. This approach excels in effectively managing and integrating spatial and temporal information, maintaining linear computational complexity, and achieving high accuracy and robustness. This method offers a novel technical solution to advance remote sensing image analysis.

While FSDFormer demonstrates significant advancements in STF, several directions warrant further investigation. Limited by computational resources, the current patch-based inference strategy introduces block artifacts that could be reduced by advanced boundary-aware training strategies or hierarchical feature blending techniques. Furthermore, the spectral fidelity of the model could be improved by applying cross-band attention mechanisms or auxiliary spectral preservation constraints. Beyond architectural enhancements, the integration of multisensor data fusion, particularly for heterogeneous sensors like optical and SAR, presents both challenges and opportunities in aligning cross-modal spatiotemporal features while preserving domain-specific characteristics. These advances aim to bridge the gap between theoretical innovation and practical deployment in real-world remote sensing scenarios.

References

[1] P. Marques, L. Pádua, J. J. Sousa, and A. Fernandes-Silva, "Advancements in remote sensing imagery applications for precision management in olive growing: A systematic review," *Remote Sens.*, vol. 16, no. 8, p. 1324, Apr. 2024.

[2] J. Xiao, A. K. Aggarwal, N. H. Duc, A. Arya, U. K. Rage, and R. Avtar, "A review of remote sensing image spatiotemporal fusion: Challenges, applications and recent trends," *Remote Sens. Appl., Soc. Environ.*, vol. 32, Nov. 2023, Art. no. 101005.

[3] M. Tshanga M, L. Ncube, and E. van Niekerk, "Remote sensing insights into subsurface-surface relationships: Land cover analysis and copper deposits exploration," *Earth Sci. Informat.*, vol. 17, no. 5, pp. 3979–4000, Oct. 2024.

[4] T. Blaschke, "Object based image analysis for remote sensing," *ISPRS J. Photogramm. Remote Sens.*, vol. 65, no. 1, pp. 2–16, Jan. 2010.

[5] H. I. Chaminé, A. J. S. C. Pereira, A. C. Teodoro, and J. Teixeira, "Remote sensing and GIS applications in Earth and environmental systems sciences," *Social Netw. Appl. Sci.*, vol. 3, no. 12, p. 870, Dec. 2021.

[6] M. Weiss, F. Jacob, and G. Duveiller, "Remote sensing for agricultural applications: A meta-review," *Remote Sens. Environ.*, vol. 236, Jan. 2020, Art. no. 111402.

[7] L. Zeng, B. D. Wardlow, D. Xiang, S. Hu, and D. Li, "A review of vegetation phenological metrics extraction using time-series, multispectral satellite data," *Remote Sens. Environ.*, vol. 237, Feb. 2020, Art. no. 111511.

[8] N. Kadhim, M. Mourshed, and M. Bray, "Advances in remote sensing applications for urban sustainability," *Euro-Mediterranean J. Environ. Integr.*, vol. 1, no. 1, p. 7, Dec. 2016.

[9] K. Wei, J. Dai, D. Hong, and Y. Ye, "MGFNet: An MLP-dominated gated fusion network for semantic segmentation of high-resolution multimodal remote sensing images," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 135, Dec. 2024, Art. no. 104241.

[10] Z. Gong, W. Ge, J. Guo, and J. Liu, "Satellite remote sensing of vegetation phenology: Progress, challenges, and opportunities," *ISPRS J. Photogramm. Remote Sens.*, vol. 217, pp. 149–164, Nov. 2024.

[11] D. Cui, S. Wang, C. Zhao, and H. Zhang, "A novel remote sensing spatiotemporal data fusion framework based on the combination of deep-learning downscaling and traditional fusion algorithm," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 7957–7970, 2024.

[12] J. Anandakrishnan, V. M. Sundaram, and P. Paneer, "STA-AgriNet: A spatio-temporal attention framework for crop type mapping from fused multi-sensor multi-temporal SITS," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 1817–1826, 2025.

[13] J. Anandakrishnan, V. M. Sundaram, and P. Paneer, "CERMF-Net: A SAR-optical feature fusion for cloud elimination from Sentinel-2 imagery using residual multiscale dilated network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 11741–11749, 2024.

[14] M. Wang, X. Zhao, and L. Ren, "Enhancing cloud-removed regions in multispectral optical images using SAR edge features," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, 2025, Art. no. 5002205.

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[15] W. Shi, D. Guo, and H. Zhang, "A reliable and adaptive spatiotemporal data fusion method for blending multi-spatiotemporal-resolution satellite images," *Remote Sens. Environ.*, vol. 268, Jan. 2022, Art. no. 112770.

[16] X. Zhu, F. Cai, J. Tian, and T. Williams, "Spatiotemporal fusion of multisource remote sensing data: Literature survey, taxonomy, principles, applications, and future directions," *Remote Sens.*, vol. 10, no. 4, p. 527, Mar. 2018.

[17] Z. Tan, L. Di, M. Zhang, L. Guo, and M. Gao, "An enhanced deep convolutional model for spatiotemporal image fusion," *Remote Sens.*, vol. 11, no. 24, p. 2898, Dec. 2019.

[18] F. Bastani, P. Wolters, R. Gupta, J. Ferdinando, and A. Kembhavi, "SatlasPretrain: A large-scale dataset for remote sensing image understanding," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 16726–16736.

[19] D. Guo and W. Shi, "Object-level hybrid spatiotemporal fusion: Reaching a better tradeoff among spectral accuracy, spatial accuracy, and efficiency," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 8007–8021, 2023.

[20] T. Benzenati, A. Kallel, and Y. Kessentini, "STF-trans: A two-stream spatiotemporal fusion transformer for very high resolution satellites images," *Neurocomputing*, vol. 563, Jan. 2024, Art. no. 126868.

[21] L. Han et al., "Using spatiotemporal fusion algorithms to fill in potentially absent satellite images for calculating soil salinity: A feasibility study," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 111, Jul. 2022, Art. no. 102839.

[22] F. Gao, J. Masek, M. Schwaller, and F. Hall, "On the blending of the Landsat and MODIS surface reflectance: Predicting daily Landsat surface reflectance," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 8, pp. 2207–2218, Aug. 2006.

[23] X. Zhu, J. Chen, F. Gao, X. Chen, and J. G. Masek, "An enhanced spatial and temporal adaptive reflectance fusion model for complex heterogeneous regions," *Remote Sens. Environ.*, vol. 114, no. 11, pp. 2610–2623, Nov. 2010.

[24] Y. Zhao, B. Huang, and H. Song, "A robust adaptive spatial and temporal image fusion model for complex land surface changes," *Remote Sens. Environ.*, vol. 208, pp. 42–62, Apr. 2018.

[25] B. Zhukov, D. Oertel, F. Lanzl, and G. Reinhackel, "Unmixing-based multisensor multiresolution image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 37, no. 3, pp. 1212–1226, May 1999.

[26] R. Zurita-Milla, J. Clevers, and M. E. Schaepman, "Unmixing-based Landsat TM and MERIS FR data fusion," *IEEE Geosci. Remote Sens. Lett.*, vol. 5, no. 3, pp. 453–457, Jul. 2008.

[27] A. Li, Y. Bo, Y. Zhu, P. Guo, J. Bi, and Y. He, "Blending multi-resolution satellite sea surface temperature (SST) products using Bayesian maximum entropy method," *Remote Sens. Environ.*, vol. 135, pp. 52–63, Aug. 2013.

[28] H. Shen, X. Meng, and L. Zhang, "An integrated framework for the spatio-temporal–spectral fusion of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 7135–7148, Dec. 2016.

[29] X. Zhu, E. H. Helmer, F. Gao, D. Liu, J. Chen, and M. A. Lefsky, "A flexible spatiotemporal method for fusing satellite images with different resolutions," *Remote Sens. Environ.*, vol. 172, pp. 165–177, Jan. 2016.

[30] K. Peng, Q. Wang, Y. Tang, X. Tong, and P. M. Atkinson, "Geographically weighted spatial unmixing for spatiotemporal fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5404217.

[31] Q. Wang and P. M. Atkinson, "Spatio-temporal fusion for daily Sentinel-2 images," *Remote Sens. Environ.*, vol. 204, pp. 31–42, Jan. 2018.

[32] X. Liu, C. Deng, J. Chanussot, D. Hong, and B. Zhao, "StfNet: A two-stream convolutional neural network for spatiotemporal image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6552–6564, Sep. 2019.

[33] Z. Tan, M. Gao, X. Li, and L. Jiang, "A flexible reference-insensitive spatiotemporal fusion model for remote sensing images using conditional generative adversarial network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5601413.

[34] J. Chen, L. Wang, R. Feng, P. Liu, W. Han, and X. Chen, "CycleGAN-STF: Spatiotemporal fusion via CycleGAN-based image generation," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5851–5865, Jul. 2021.

[35] H. Huang, W. He, H. Zhang, Y. Xia, and L. Zhang, "STFDiff: Remote sensing image spatiotemporal fusion with diffusion models," *Inf. Fusion*, vol. 111, Nov. 2024, Art. no. 102505.

[36] W. Li, D. Cao, Y. Peng, and C. Yang, "MSNet: A multi-stream fusion network for remote sensing spatiotemporal fusion based on transformer and convolution," *Remote Sens.*, vol. 13, no. 18, p. 3724, Sep. 2021.

[37] G. Chen, P. Jiao, Q. Hu, L. Xiao, and Z. Ye, "SwinSTFM: Remote sensing spatiotemporal fusion using Swin Transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5410618.

[38] S. Jamil, M. Jalil Piran, and O.-J. Kwon, "A comprehensive survey of transformers for computer vision," *Drones*, vol. 7, no. 5, p. 287, Apr. 2023.

[39] X. He, Y. Zhou, J. Zhao, D. Zhang, R. Yao, and Y. Xue, "Swin transformer embedding UNet for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4408715.

[40] L. Gao et al., "STransFuse: Fusing Swin transformer and convolutional neural network for remote sensing image semantic segmentation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 10990–11003, 2021.

[41] L. Wang et al., "UNetFormer: A UNet-like transformer for efficient semantic segmentation of remote sensing urban scene imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 190, pp. 196–214, Aug. 2022.

[42] C. Zhang, J. Su, Y. Ju, K.-M. Lam, and Q. Wang, "Efficient inductive vision transformer for oriented object detection in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5616320.

[43] X. Jiang and Y. Wu, "Remote sensing object detection based on convolution and Swin transformer," *IEEE Access*, vol. 11, pp. 38643–38656, 2023.

[44] S. K. Roy, A. Deria, D. Hong, B. Rasti, A. Plaza, and J. Chanussot, "Multimodal fusion transformer for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5515620.

[45] X. He, Y. Chen, and Z. Lin, "Spatial–spectral transformer for hyperspectral image classification," *Remote Sens.*, vol. 13, no. 3, p. 498, Jan. 2021.

[46] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 5998–6008.

[47] S. Zhuoran, Z. Mingyuan, Z. Haiyu, Y. Shuai, and L. Hongsheng, "Efficient attention: Attention with linear complexities," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2021, pp. 3531–3539.

[48] D. Han et al., "Agent attention: On the integration of softmax and linear attention," in *Proc. Eur. Conf. Comput. Vis.*, 2024, pp. 124–140.

[49] D. Han, X. Pan, Y. Han, S. Song, and G. Huang, "FLatten transformer: Vision transformer using focused linear attention," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 5938–5948.

[50] J. Li, Y. Li, L. He, J. Chen, and A. Plaza, "Spatio-temporal fusion for remote sensing data: An overview and new benchmark," *Sci. China Inf. Sci.*, vol. 63, no. 4, Apr. 2020, Art. no. 140301.

[51] I. V. Emelyanova, T. R. McVicar, T. G. Van Niel, L. T. Li, and A. I. J. M. van Dijk, "Assessing the accuracy of blending Landsat–MODIS surface reflectances in two landscapes with contrasting spatial and temporal dynamics: A framework for algorithm selection," *Remote Sens. Environ.*, vol. 133, pp. 193–209, Jun. 2013.

[52] H. Guo, D. Ye, H. Xu, and L. Bruzzone, "OBSUM: An object-based spatial unmixing model for spatiotemporal fusion of remote sensing images," *Remote Sens. Environ.*, vol. 304, Apr. 2024, Art. no. 114046.

[53] X. Zhang, S. Li, Z. Tan, and X. Li, "Enhanced wavelet based spatiotemporal fusion networks using cross-paired remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 211, pp. 281–297, May 2024.

[54] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004.

[55] S. Kaneko, Y. Satoh, and S. Igarashi, "Using selective correlation coefficient for robust image registration," *Pattern Recognit.*, vol. 36, no. 5, pp. 1165–1173, May 2003.

[56] R. Yuhas, A. Goetz, and J. Boardman, "Discrimination among semi-arid landscape endmembers using the spectral angle mapper (sam) algorithm," in *Proc. Summaries Annu. JPL Airborne Geosci. Workshop*, 1992, pp. 147–149.

[57] M. M. Khan, L. Alparone, and J. Chanussot, "Pansharpening quality assessment using the modulation transfer functions of instruments," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 11, pp. 3880–3891, Nov. 2009.

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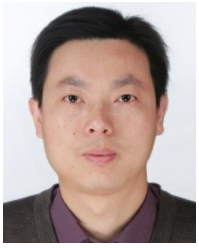
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